gcddomain

1 Base

Definition. The elements of Base_Top are a basic type.

Definition. Base_Top_tt is an element of Base_Top.

Definition. The elements of Base_Empty are a basic type.

Definition. The elements of Bool are a basic type.

Definition. true is an element of Bool.

Definition. We can say that we have a contradiction.

Definition. The elements of Base_Nat are a basic type.

Definition. Base_Nat_zero is an element of Base_Nat.

Definition. Let n be an element of Base_Nat. Then Base_Nat_suc applied to n is an element of Base_Nat.

Definition. Let x and x_1 be elements of Bool. Then we can say that x and x_1.

Definition. Let x and x_1 be elements of Bool. Then we can say that x or x_1.

Definition. Let x be an element of Bool. Then we can say that it is not the case that x.

Definition. Let x and x_1 be elements of Base_Nat. Then leb applied to x and x_1 is an element of Bool.

Definition. Let x and x_1 be elements of Base_Nat. Then eqN applied to x and x_1 is an element of Bool.

IsTop. Let u and v be elements of Base_Top. Then $u = v$.

2 Prelude

Definition. Let A be a set. Assume that we have a contradiction. Then Prelude_absurd applied to A and x is an element of A.

Prelude_sym. Let A be a set. Let x and y be elements of A such that $x = y$. Then $y = x$.

Prelude_trans. Let A be a set. Let x, y and z be elements of A such that $x = y$ and $y = z$. Then $x = z$.

Prelude_ap. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let x and y be elements of A such that $x = y$. Then $f(x) = f(y)$.

Cong2. Let B be a set. Let op be a function from elements x of B and elements x_1 of B to elements of B. Let a, a_1, b and b_1 be elements of B such that $a = a_1$ and $b = b_1$. Then $\mathrm{op}(a, b) = \mathrm{op}(a_1, b_1)$.

Definition. Let A be a set. Let P be a function from elements x of A to sets. Let x and y be elements of A such that $x = y$. Let u be an element of P applied to x. Then Prelude_subst applied to A, x, P, x, y, p and u is an element of P applied to y.

Definition. Let A and B be sets such that $A = B$. Let a be an element of A. Then coe applied to A, B, p and a is an element of B.

Happly. Let A be a set. Let B be a function from elements x of A to sets. Let f and g be functions from elements x of A to elements of B applied to x such that $f = g$. Let x be an element of A. Then $f(x) = g(x)$.

Trans_refl_l. Let A be a set. Let x and y be elements of A such that $x = y$. Then $\mathrm{Prelude_trans}(A, x, x, y, \mathrm{refl}(A, x), p) = p$.

Trans_sym_l. Let A be a set. Let x and y be elements of A such that $x = y$. Then $\mathrm{Prelude_trans}(A, y, x, y, \mathrm{Prelude_sym}(A, x, y, p), p) = \mathrm{refl}(A, y)$.

Trans_cancel. Let A be a set. Let x, y and z be elements of A such that $x = y$ and $y = z$. Then $\mathrm{Prelude_trans}(A, y, x, z, \mathrm{Prelude_sym}(A, x, y, a), \mathrm{Prelude_trans}(A, x, y, z, a, p)) = p$.

SymU. Let A and B be sets such that $A = B$. Then $B = A$.

Coe_sym. Let A and B be sets such that $A = B$. Let b be an element of B. Then $\mathrm{coe}(A, B, p, \mathrm{coe}(B, A, \mathrm{symU}(A, B, p), b)) = b$.

Definition. Let A be a set. Then A is a proposition, if for all elements x and y of A, $x = y$.

Definition. Let A be a set. Then A is a set, if for all elements x and y of A, $x = y$ is a proposition.

Retract_prop. Let A and B be sets. Let s be a function from elements x of A to elements of B. Let r be a function from elements x of B to elements of A such that for all elements a of A, $r(s(a)) = a$. Assume that B is a proposition. Then A is a proposition.

Sigma_eq. Let A be a set. Let B be a function from elements x of A to sets. Let a and a' be elements of A. Let b be an element of B applied to a. Let b' be an element of B applied to a'. Assume that $a = \mathrm{a}'$. Assume that B applied to a' is a proposition. Then $(a, b) = (\mathrm{a}', \mathrm{b}')$.

IsProp_Sigma. Let A be a set. Let B be a function from elements x of A to sets. Assume that A is a proposition. Assume that for all elements a of A, B applied to a is a proposition. Then SigmaType applied to A and the function that maps x to B applied to x is a proposition.

IsProp_Empty. Base_Empty is a proposition.

2.1 Prelude.Hedberg

In this section: Let A be a set. Let x be an element of A. Let R be a function from elements x_1 of A to sets such that for all elements y of A, R applied to y is a proposition. Let r be an element of R applied to x. Assume that for all elements y of A, for all elements x_1 of R applied to y, $x = y$.

Fx. $x = x$.

Canon. Let y be an element of A such that $x = y$. Then $\mathrm{Prelude_trans}(A, x, x, y, \mathrm{fx}(A, x, R, \mathrm{Rprop}, r, f), p) = f(y, \mathrm{Prelude_subst}(A, R, x, y, p, r))$.

Ts. Let y be an element of A such that $x = y$. Then $x = y$.

Definition. Let y be an element of A such that $x = y$. Then $\mathrm{norm}(y, p)$ is an element of Proof applied to $p = \mathrm{ts}(A, x, R, \mathrm{Rprop}, r, f, y, f(y, \mathrm{Prelude_subst}(A, R, x, y, p, r)))$.

Hedberg. Let y be an element of A. Then $x = y$ is a proposition.

3 Prelude

Prop_to_set. Let A be a set such that A is a proposition. Then A is a set.

IsEquiv. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let sec and ret be functions from elements x of B to elements of A such that for all elements b of B, $f(\mathrm{sec}(b)) = b$ and for all elements a of A, $\mathrm{ret}(f(a)) = a$. Then Structure.

Definition. Let A and B be sets. Let f be a function from elements x of A to

elements of B. Let sec and ret be functions from elements x of B to elements of A such that for all elements b of B, $f(\mathrm{sec}(b)) = b$ and for all elements a of A, $\mathrm{ret}(f(a)) = a$. Then IsEquiv[A, B, f]_mk applied to A, B, x, f, x, sec, sec $-$ eq, x, ret and ret $-$ eq is an element of IsEquiv[A, B, f] applied to A, B and f.

Definition. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let t be an element of IsEquiv[A, B, f] applied to A, B and f. Let x be an element of B. Then IsEquiv[A, B, f]_sec applied to A, B, x, f, t and x is an element of A.

IsEquiv[A,B,f]_sec_eq. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let t be an element of IsEquiv[A, B, f] applied to A, B and f. Let b be an element of B. Then $f(\mathrm{IsEquiv[A, B, f]_sec}(A, B, f, t, b)) = b$.

Definition. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let t be an element of IsEquiv[A, B, f] applied to A, B and f. Let x be an element of B. Then IsEquiv[A, B, f]_ret applied to A, B, x, f, t and x is an element of A.

IsEquiv[A,B,f]_ret_eq. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let t be an element of IsEquiv[A, B, f] applied to A, B and f. Let a be an element of A. Then $\mathrm{IsEquiv[A, B, f]_ret}(A, B, f, t, f(a)) = a$.

Definition. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let sec and ret be functions from elements x of B to elements of A such that for all elements b of B, $f(\mathrm{sec}(b)) = b$ and for all elements a of A, $\mathrm{ret}(f(a)) = a$. Then mkIsEquiv applied to A, B, x, f, x, sec, sec $-$ eq, x, ret and ret $-$ eq is an element of IsEquiv.

Definition. Let A and B be sets. Let f be a function from elements x of A to elements of B. Then an element of isEquiv applied to A, B, x and f is an element of the function that maps k, k_1, k_2 and f to IsEquiv[A, B, f] applied to A, B, x and f.

Sec_ret. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let e be an element of isEquiv applied to A, B and f. Let a be an element of A. Then $\mathrm{IsEquiv[A, B, f]_sec}(A, B, f, e, f(a)) = a$.

Definition. Let A and B be sets. Then an element of Equiv applied to A and B is an element of the function that maps A and B to SigmaType applied to PiType applied to A and the function that maps x to B and the function that maps f to isEquiv applied to A, B and f.

Definition. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let g be a function from elements x of B to elements of A such that for all elements b of B, $f(g(b)) = b$ and for all elements a of A, $g(f(a)) = a$. Then qinv applied to A, B, x, f, x, g, fg and gf is an element of isEquiv applied to A, B and f.

Definition. Let A be a set. Then idEquiv applied to A is an element of Equiv applied to A and A.

IsoData. Let A and B be sets. Let fun be a function from elements x of A to elements of B. Let inv be a function from elements x of B to elements of A such that for all elements b of B, $\mathrm{fun}(\mathrm{inv}(b)) = b$ and for all elements a of A, $\mathrm{inv}(\mathrm{fun}(a)) = a$. Then Structure.

Definition. Let A and B be sets. Let fun be a function from elements x of A to elements of B. Let inv be a function from elements x of B to elements of A such that for all elements b of B, $\mathrm{fun}(\mathrm{inv}(b)) = b$ and for all elements a of A, $\mathrm{inv}(\mathrm{fun}(a)) = a$. Then IsoData[A, B]_mk applied to A, B, x, fun, x, inv, $\mathrm{fun} - \mathrm{inv}$ and $\mathrm{inv} - \mathrm{fun}$ is an element of IsoData[A, B] applied to A and B.

Definition. Let A and B be sets. Let t be an element of IsoData[A, B] applied to A and B. Let x be an element of A. Then IsoData[A, B]_fun applied to A, B, t and x is an element of B.

Definition. Let A and B be sets. Let t be an element of IsoData[A, B] applied to A and B. Let x be an element of B. Then IsoData[A, B]_inv applied to A, B, t and x is an element of A.

IsoData[A,B]_fun_inv. Let A and B be sets. Let t be an element of IsoData[A, B] applied to A and B. Let b be an element of B. Then IsoData[A, B]_fun$(A, B, t,$ IsoData[A, B]_inv$(A, B, t, b))$ = b.

IsoData[A,B]_inv_fun. Let A and B be sets. Let t be an element of IsoData[A, B] applied to A and B. Let a be an element of A. Then IsoData[A, B]_inv$(A, B, t,$ IsoData[A, B]_fun$(A, B, t, a))$ = a.

Definition. Let A and B be sets. Let fun be a function from elements x of A to elements of B. Let inv be a function from elements x of B to elements of A such that for all elements b of B, $\mathrm{fun}(\mathrm{inv}(b)) = b$ and for all elements a of A, $\mathrm{inv}(\mathrm{fun}(a)) = a$. Then mkIsoData applied to A, B, x, fun, x, inv, $\mathrm{fun} - \mathrm{inv}$ and $\mathrm{inv} - \mathrm{fun}$ is an element of IsoData.

Definition. Let A and B be sets. Then an element of Iso applied to A and B is an element of the function that maps k and k_1 to IsoData[A, B] applied to A and B.

Sigma_path. Let A be a set. Let B be a function from elements x of A to sets. Let a and a' be elements of A. Let b be an element of B applied to a. Let b' be an element of B applied to a'. Assume that $a = \mathrm{a}'$. Assume that Prelude_subst$(A, B, a, \mathrm{a}', p, b) = \mathrm{b}'$. Then $(a, b) = (\mathrm{a}', \mathrm{b}')$.

SPath. Let A be a set. Let B be a function from elements x of A to sets. Let u and v be elements of SigmaType applied to A and the function that maps x to B applied to x such that $u = v$. Then there exists an element p of $\pi_1(u) = \pi_1(v)$, such that Prelude_subst$(A, B, \pi_1(u), \pi_1(v), p, \pi_2(u)) = \pi_2(v)$.

RsPath. Let A be a set. Let B be a function from elements x of A to sets. Let u and v be elements of SigmaType applied to A and the function that maps x to B applied to x such that $u = v$. Then Sigma_path$(A, B, \pi_1(u), \pi_1(v), \pi_2(u), \pi_2(v), \pi_1(\text{sPath}(A, B, u, v, P)), \pi_2($
P.

IsSet_Sigma. Let A be a set. Let B be a function from elements x of A to sets. Assume that A is a set. Assume that for all elements a of A, B applied to a is a set. Then SigmaType applied to A and the function that maps x to B applied to x is a set.

IsSet_retract. Let A and B be sets. Let s be a function from elements x of A to elements of B. Let r be a function from elements x of B to elements of A such that for all elements a of A, $r(s(a)) = a$. Assume that B is a set. Then A is a set.

Loop_refl. Let A be a set such that A is a set. Let x be an element of A such that $x = x$. Then $q = \text{refl}(A, x)$.

Subst_proj. Let A and X be sets. Let g be a function from elements x of A to elements of X. Let F be a function from elements x of A to sets. Let H be a function from elements x of X to sets. Let π be a function from elements a of A and elements x of F applied to a to elements of H applied to g applied to a. Let a and a' be elements of A such that $a = \mathrm{a}'$. Let u be an element of F applied to a. Then $\pi(\mathrm{a}', \text{Prelude_subst}(A, F, a, \mathrm{a}', P, u)) = \text{Prelude_subst}(X, H, g(a), g(\mathrm{a}'), \text{Prelude_ap}(A, X, g, a, \mathrm{a}', P), \pi(a, u))$.

Subst_ap. Let A and X be sets. Let g be a function from elements x of A to elements of X. Let H be a function from elements x of X to sets. Let a and a' be elements of A such that $a = \mathrm{a}'$. Let u be an element of H applied to g applied to a. Then $\text{Prelude_subst}(A, \lambda a_1.H(g(a_1)), a, \mathrm{a}', P, u) = \text{Prelude_subst}(X, H, g(a), g(\mathrm{a}'), \text{Prelude_ap}(A, X, g, a, \mathrm{a}', P), u)$.

Sigma_snd. Let A be a set. Let B be a function from elements x of A to sets. Let a and a' be elements of A. Let b be an element of B applied to a. Let b' be an element of B applied to a' such that $(a, b) = (\mathrm{a}', \mathrm{b}')$. Then $\text{Prelude_subst}(A, B, a, \mathrm{a}', \text{Prelude_ap}(\text{SigmaType}(A, \lambda x.B(x)), A, \pi_1(,)(a, b), (\mathrm{a}', \mathrm{b}'), Q), b) = \mathrm{b}'$.

4 Axioms

Definition. An element of FunExt is an element of PiType applied to the set of sets and the function that maps A to PiType applied to PiType applied to A and the function that maps x to the set of sets and the function that maps B to PiType applied to PiType applied to A and the function that maps x to B applied to x and the function that maps f to PiType applied to PiType applied to A and the function that maps x to B applied to x and the function that

maps g to isEquiv applied to $f = g$, the proposition that for all elements x of A, $f(x) = g(x)$ and the function that maps k and k_1 to happly applied to A, B, f, g, k and k_1.

Definition. Let A and B be sets such that $A = B$. Then idtoeqv applied to A, B and p is an element of Equiv applied to A and B.

Definition. An element of Univalence is an element of PiType applied to the set of sets and the function that maps A to PiType applied to the set of sets and the function that maps B to isEquiv applied to $A = B$, Equiv applied to A and B and the function that maps k to idtoeqv applied to A, B and k.

Definition. An element of PropUnivalence is an element of PiType applied to the set of sets and the function that maps P to PiType applied to the set of sets and the function that maps Q to PiType applied to the proposition that P is a proposition and the function that maps x to PiType applied to the proposition that Q is a proposition and the function that maps x_1 to isEquiv applied to $P = Q$, Equiv applied to P and Q and the function that maps k to idtoeqv applied to P, Q and k.

4.1 Axioms.FunExtLemmas

In this section: Let fe be an element of FunExt.

Funext. Let A be a set. Let B be a function from elements x of A to sets. Let f and g be functions from elements x of A to elements of B applied to x such that for all elements x of A, $f(x) = g(x)$. Then $f = g$.

Happly_funext. Let A be a set. Let B be a function from elements x of A to sets. Let f and g be functions from elements x of A to elements of B applied to x such that for all elements x of A, $f(x) = g(x)$. Then $\lambda k.\mathrm{happly}(A, B, f, g, \mathrm{funext}(\mathrm{fe}, A, B, f, g, h), k) = h$.

Funext_happly. Let A be a set. Let B be a function from elements x of A to sets. Let f and g be functions from elements x of A to elements of B applied to x such that $f = g$. Then $\mathrm{funext}(\mathrm{fe}, A, B, f, g, \lambda k.\mathrm{happly}(A, B, f, g, p, k)) = p$.

IsProp_Pi. Let A be a set. Let B be a function from elements x of A to sets such that for all elements x of A, B applied to x is a proposition. Then PiType applied to A and the function that maps x to B applied to x is a proposition.

IsProp_isProp. Let A be a set. Then the proposition that A is a proposition is a proposition.

IsSet_Pi. Let A be a set. Let B be a function from elements x of A to sets such that for all elements x of A, B applied to x is a set. Then PiType applied to A and the function that maps x to B applied to x is a set.

4.2 Axioms.UnivalenceLemmas

Ua. Let A and B be sets. Let e be an element of Equiv applied to A and B. Then $A = B$.

Definition. prop_univ is an element of PropUnivalence.

Definition. Let T and T' be sets. Let P be a function from sets T' and elements x of Equiv applied to T and T' to sets. Let d be an element of P applied to T and idEquiv applied to T. Let e be an element of Equiv applied to T and T'. Then eqind applied to T, T', x, P, d, T' and e is an element of P applied to T' and e.

Fst_idtoeqv. Let A and B be sets such that $A = B$. Let a be an element of A. Then $\pi_1(\mathrm{idtoeqv}(A, B, p))a = \mathrm{coe}(A, B, p, a)$.

Ua_beta. Let A and B be sets. Let e be an element of Equiv applied to A and B. Let a be an element of A. Then $\mathrm{coe}(A, B, \mathrm{ua}(A, B, e), a) = \pi_1(e)a$.

5 FunExtUA

Definition. Let T be a set. Then isContr holds for T, if there exists an element c of T, such that for all elements y of T, $c = y$.

PathC. Let T be a set such that isContr holds for T. Let u and v be elements of T. Then $u = v$.

PathC_refl. Let T be a set such that isContr holds for T. Let u be an element of T. Then $\mathrm{pathC}(T, k, u, u) = \mathrm{refl}(T, u)$.

Sing. Let T be a set. Let t be an element of T. Then isContr holds for the proposition that there exists an element y of T, such that $y = t$.

Definition. Let X, C and D be sets. Let f be a function from elements x of C to elements of D. Let h be a function from elements x of X to elements of C. Let x be an element of X. Then post applied to X, C, D, x, f, x, h and x is an element of D.

Definition. Let X, C and D be sets. Let f be a function from elements x of C to elements of D. Let t be a function from elements x of X to elements of D. Then fibP holds for X, C, D, x, f, x and t, if there exists an element h of PiType applied to X and the function that maps x to C, such that $\lambda k.\mathrm{post}(X, C, D, x, h, k) = f$.

Cfp. Let X, C and D be sets such that $C = D$. Let t be a function from elements x of X to elements of D. Then isContr holds for fibP applied to X, C, D, $\pi_1(\mathrm{idtoeqv}(C, D, p))$ and t.

5.1 FunExtUA.FE

Cf. Let X, C and D be sets. Let e be an element of Equiv applied to C and D. Let t be a function from elements x of X to elements of D. Then isContr holds for fibP applied to X, C, D, $\pi_1(e)$ and t.

Definition. Let A be a set. Let P be a function from elements x of A to sets such that for all elements x of A, isContr holds for P applied to x. Then pr1E applied to A, x, P and c is an element of Equiv applied to SigmaType applied to A and the function that maps x to P applied to x and A.

Wfe. Let A be a set. Let P be a function from elements x of A to sets such that for all elements x of A, isContr holds for P applied to x. Then isContr holds for PiType applied to A and the function that maps x to P applied to x.

TotC. Let A be a set. Let B be a function from elements x of A to sets. Let f be a function from elements x of A to elements of B applied to x. Then isContr holds for the proposition that there exists an element g of PiType applied to A and the function that maps x to B applied to x, such that for all elements x of A, $f(x) = g(x)$.

Definition. FE_fe is an element of FunExt.

6 Replacement

Definition. Let B be a set. Then an element of Small applied to B is an element of the function that maps B to SigmaType applied to the set of sets and the function that maps X to Equiv applied to X and B.

Definition. Let B be a set. Then an element of LocallySmall applied to B is an element of the function that maps B to PiType applied to B and the function that maps x to PiType applied to B and the function that maps y to Small applied to $x = y$.

Definition. Let X and B be sets. Let i be a function from elements x of X to elements of B. Then an element of Embedding applied to X, B, x and i is an element of the function that maps X, B, i and i to PiType applied to X, the function that maps x_1 to PiType applied to X and the function that maps y to isEquiv applied to $x_1 = y$, $x(x_1) = x(y)$ and the function that maps k to Prelude_ap applied to X, B, x, x_1, y and k and i.

Image. Let A, B and carrier be sets. Let f be a function from elements x of A to elements of B. Let quotient be a function from elements x of A to elements of carrier. Let inclusion be a function from elements x of carrier to elements of B such that for all elements a of A, inclusion(quotient(a)) = $f(a)$. Let embedding be an element of Embedding applied to carrier, B and inclusion.

Then Structure.

Definition. Let A, B and carrier be sets. Let f be a function from elements x of A to elements of B. Let quotient be a function from elements x of A to elements of carrier. Let inclusion be a function from elements x of carrier to elements of B such that for all elements a of A, inclusion(quotient(a)) = $f(a)$. Let embedding be an element of Embedding applied to carrier, B and inclusion. Then mkImage applied to A, B, x, f, carrier, x, quotient, x, inclusion, triangle and embedding is an element of Image applied to A, B and f.

Definition. An element of Images is an element of PiType applied to the set of sets and the function that maps A to PiType applied to the set of sets and the function that maps B to PiType applied to LocallySmall applied to B and the function that maps x to PiType applied to PiType applied to A and the function that maps x_1 to B and the function that maps f to Image applied to A, B and f.

Definition. Let A and X be sets. Let q be a function from elements x of A to elements of X. Let F be a function from elements x of X to sets. Let x be a function from elements x of X to elements of F applied to x. Let a be an element of A. Then precompose applied to A, X, x, q, x, F, x, x and a is an element of F applied to q applied to a.

Definition. An element of Surjectivity is an element of PiType applied to the set of sets and the function that maps A to PiType applied to the set of sets and the function that maps B to PiType applied to LocallySmall applied to B and the function that maps small to PiType applied to PiType applied to A and the function that maps x to B and the function that maps f to PiType applied to PiType applied to Image_carrier applied to A, B, f and images and the function that maps x to the set of sets and the function that maps F to PiType applied to the proposition that for all elements x of Image_carrier applied to A, B, f and images, F applied to x is a proposition and the function that maps prop to isEquiv applied to PiType applied to Image_carrier applied to A, B, f and images and the function that maps x to F applied to x, PiType applied to A and the function that maps a to F applied to quotient applied to A, B, f, images and a and the function that maps k′ and k_1 to precompose applied to A, Image_carrier applied to A, B, f and images, the function that maps k to quotient applied to A, B, f, images and k, F, k′ and k_1.

7 Ring

Definition. The elements of Ring_Top are a basic type.

Definition. Ring_Top_tt is an element of Ring_Top.

Definition. $\emptyset$ is a set.

Definition. $\mathbb{N}$ is a set.

Definition. 0 is a natural number.

Definition. Let n be a natural number. Then $n + 1$ is a natural number.

Definition. Let A and B be sets. Then the elements of Plus applied to A and B are a basic type.

Definition. Let A and B be sets. Let x be an element of A. Then inl applied to A, B and x is an element of Plus applied to A and B.

Definition. Let A and B be sets. Let y be an element of B. Then inr applied to A, B and y is an element of Plus applied to A and B.

Definition. Let A be a set. Then the lists of elements of A are a basic type.

Definition. Let A be a set. Then 0 is a list of elements of A.

Definition. Let A be a set. Let x be an element of A. Let xs be a list of elements of A. Then $x + X$xs is a list of elements of A.

Definition. Let A be a set. Assume that we have a contradiction. Then Ring_absurd applied to A and x is an element of A.

Definition. Let A be a set. Then $\neg A$ is a proposition defined as the proposition that for all elements x of A, we have a contradiction.

Definition. Let A be a set. Assume that we have a contradiction. Then efq applied to A and e is an element of A.

Ring_sym. Let A be a set. Let x and y be elements of A such that $x = y$. Then $y = x$.

Ring_trans. Let A be a set. Let x, y and w be elements of A such that $x = y$ and $y = w$. Then $x = w$.

Ring_ap. Let A and B be sets. Let f be a function from elements x of A to elements of B. Let x and y be elements of A such that $x = y$. Then $f(x) = f(y)$.

Definition. Let A be a set. Let P be a function from elements x of A to sets. Let x and y be elements of A such that $x = y$. Let u be an element of P applied to x. Then Ring_subst applied to A, x, P, x, y, p and u is an element of P applied to y.

Definition. Let x and x_1 be natural numbers. Then $x + x_1$ is a natural number.

Definition. Let x be a natural number. Then pred applied to x is a natural number.

Definition. Let x be a natural number. Then the elements of isZero applied to x are a basic type.

Zero_suc. Let n be a natural number such that $0 = n + 1$. Then we have a contradiction.

Suc_zero. Let n be a natural number such that $n + 1 = 0$. Then we have a contradiction.

Suc_inj. Let m and n be natural numbers such that $m+1 = n+1$. Then $m = n$.

Plus_suc. Let m and n be natural numbers. Then $m + n + 1 = (m + n) + 1$.

Plus_zero. Let m be a natural number. Then $m + 0 = m$.

Plus_comm. Let m and n be natural numbers. Then $m + n = n + m$.

Definition. Let A be a set. Let x be a list of elements of A. Then len applied to A and x is a natural number.

Definition. A commutative ring is an algebraic structure, where

- R is a set.
- z is an element of R.
- o is an element of R.
- $+$ is a binary operation on R.
- $\cdot$ is a binary operation on R.
- neg is a function from elements x of R to elements of R.
- for all elements x, y and w of R, $x + (y + w) = (x + y) + w$.
- for all elements x and y of R, $x + y = y + x$.
- for all elements x of R, $x + z = x$.
- for all elements x of R, $x + \text{neg}(x) = z$.
- for all elements x, y and w of R, $x \cdot (y \cdot w) = (x \cdot y) \cdot w$.
- for all elements x and y of R, $x \cdot y = y \cdot x$.
- for all elements x of R, $x \cdot o = x$.
- for all elements x, y and w of R, $x \cdot (y + w) = (x \cdot y) + (x \cdot w)$.

Definition. Let R be a set. Let z and o be elements of R. Let add, mul and neg be functions from elements x of R and elements x_1 of R to elements of R such that for all elements x of R, for all elements y of R, for all elements w of R, $\text{add}(x, \text{add}(y, w)) = \text{add}(\text{add}(x, y), w)$, for all elements x of R, for all elements y of R, $\text{add}(x, y) = \text{add}(y, x)$, for all elements x of R, $\text{add}(x, z) = x$, for all

elements x of R, $\mathrm{add}(x, \mathrm{neg}(x)) = z$, for all elements x of R, for all elements y of R, for all elements w of R, $\mathrm{mul}(x, \mathrm{mul}(y, w)) = \mathrm{mul}(\mathrm{mul}(x, y), w)$, for all elements x of R, for all elements y of R, $\mathrm{mul}(x, y) = \mathrm{mul}(y, x)$, for all elements x of R, $\mathrm{mul}(x, o) = x$ and for all elements x of R, for all elements y of R, for all elements w of R, $\mathrm{mul}(x, \mathrm{add}(y, w)) = \mathrm{add}(\mathrm{mul}(x, y), \mathrm{mul}(x, w))$. Then mkCRing applied to R, z, o, x, x_1, add, x, x_1, mul, x, neg, addA, addC, addZ, addN, mulA, mulC, mulO and dist is a commutative ring.

7.1 Ring.RL

In this section: Let Rg be a commutative ring.

Definition. Let x be an element of R. Then unit holds for x, if there exists an element y of R, such that $xy = 1$.

Zl. Let x be an element of R. Then $0 + x = x$.

Add_cancel. Let u and v be elements of R such that $u + v = u$. Then $v = 0$.

Neg_uniq. Let x and y be elements of R such that $x + y = 0$. Then $y = -x$.

Mul_zero. Let x be an element of R. Then $x0 = 0$.

Neg_zero. $-0 = 0$.

Zero_negmul. Let x be an element of R. Then $0 + -(x0) = 0$.

Add_neg_cancel. Let x and y be elements of R. Then $(x + y) + -y = x$.

Add_mulzero. Let x and y be elements of R. Then $x + (y0) = x$.

Zero_mul. Let x be an element of R. Then $0x = 0$.

Unit_reg. Let u and v be elements of R such that unit holds for Rg and u and $uv = 0$. Then $v = 0$.

Mul_neg. Let x and y be elements of R. Then $x - y = -(xy)$.

RL_swap. Let p, q, r and s be elements of R. Then $(p + q) + (r + s) = (p + r) + (q + s)$.

Negsub. Let x and y be elements of R. Then $-(x + -y) = -x + y$.

Expand. Let a_1, b, v and w be elements of R. Then $-(a_1(v + -(bw))) = -(a_1 v) + (a_1(bw))$.

Mul_swap. Let a_1, b and w be elements of R. Then $a_1(bw) = b(a_1 w)$.

One_left. Let x be an element of R. Then $1x = x$.

Add_swap. Let x, y and w be elements of R. Then $x + (y + w) = y + (x + w)$.

Dist_left. Let x, y and w be elements of R. Then $(x + y)w = (xw) + (yw)$.

Neg_cancel. Let x be an element of R. Then $-x + x = 0$.

Neg_neg. Let x be an element of R. Then $- - x = x$.

Neg_add. Let x and y be elements of R. Then $-(x + y) = -x + -y$.

Neg_mul_left. Let x and y be elements of R. Then $-xy = -(xy)$.

Mul_four. Let a_1, b, c and d be elements of R. Then $(a_1 b)(cd) = (a_1 c)(bd)$.

Definition. Let a_1, b, u, v and w be elements of R. Then H_2 applied to a_1, b, u, v and w is an element of R.

H2_comm. Let a_1, b, u, v and w be elements of R. Then $H_2(\mathrm{Rg}, a_1, b, u, v, w) = H_2(\mathrm{Rg}, b, a_1, u, v, w)$.

7.2 Poly.PolyR

In this section: Let Rg be a commutative ring.

Definition. Let Rg be a commutative ring. Then $\mathcal{P}$ is a set defined as the set of lists of elements of R.

Definition. Let Rg be a commutative ring. Let p be an element of $\mathcal{P}$. Then $|p|$ is a natural number.

Definition. Let Rg be a commutative ring. Let x be an element of $\mathcal{P}$. Let x_1 be a natural number. Then x_{x_1} is an element of R.

Definition. Let Rg be a commutative ring. Let p and q be elements of $\mathcal{P}$. Then p is equivalent to q, if for all natural numbers k, $p_k = q_k$.

Peq_sym. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Then $q \sim p$.

Peq_trans. Let p, q and s be elements of $\mathcal{P}$ such that $p \sim q$ and $q \sim s$. Then $p \sim s$.

Definition. Let x be an element of $\mathcal{P}$. Let x_1 be an element of R. Then ev applied to x and x_1 is an element of R.

Definition. Let q be an element of $\mathcal{P}$. Let x be a natural number. Then cprev applied to q and x is an element of R.

Cprev_peq. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Let k be a natural number. Then $\mathrm{cprev}(\mathrm{Rg}, p, k) = \mathrm{cprev}(\mathrm{Rg}, q, k)$.

Definition. Let b be an element of R. Let x be an element of $\mathcal{P}$. Then addc applied to b and x is an element of $\mathcal{P}$.

Addc0. Let b be an element of R. Let p be an element of $\mathcal{P}$. Then $\mathrm{addc}(\mathrm{Rg}, b, p)_0 = b + (p_0)$.

AddcS. Let b be an element of R. Let p be an element of $\mathcal{P}$. Let k be a natural number. Then $\mathrm{addc}(\mathrm{Rg}, b, p)_{k+1} = p_{k+1}$.

Addcz. Let p be an element of $\mathcal{P}$. Then $\mathrm{addc}(\mathrm{Rg}, 0, p) \sim p$.

Definition. Let a_1 be an element of R. Let x be an element of $\mathcal{P}$. Then mulXa applied to a_1 and x is an element of $\mathcal{P}$.

Definition. Let a_1 be an element of R. Let q be an element of $\mathcal{P}$. Let k be a natural number. Then mform applied to a_1, q and k is an element of R.

MulXaS. Let a_1 and b be elements of R. Let q be an element of $\mathcal{P}$ such that for all natural numbers k, $\mathrm{mulXa}(\mathrm{Rg}, a_1, q)_k = \mathrm{mform}(\mathrm{Rg}, a_1, q, k)$. Let k be a natural number. Then $\mathrm{addc}(\mathrm{Rg}, b, \mathrm{mulXa}(\mathrm{Rg}, a_1, q))_k = \mathrm{mform}(\mathrm{Rg}, a_1, b + Xq, k + 1)$.

Coef_mulXa. Let a_1 be an element of R. Let q be an element of $\mathcal{P}$. Let k be a natural number. Then $\mathrm{mulXa}(\mathrm{Rg}, a_1, q)_k = \mathrm{mform}(\mathrm{Rg}, a_1, q, k)$.

Mform_peq. Let a_1 be an element of R. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Let k be a natural number. Then $\mathrm{mform}(\mathrm{Rg}, a_1, p, k) = \mathrm{mform}(\mathrm{Rg}, a_1, q, k)$.

MulXa_peq. Let a_1 be an element of R. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Then $\mathrm{mulXa}(\mathrm{Rg}, a_1, p) \sim \mathrm{mulXa}(\mathrm{Rg}, a_1, q)$.

Definition. Let al be an element of R. Let x be an element of $\mathcal{P}$. Then prodL applied to al and x is an element of $\mathcal{P}$.

Definition. Let x and x_2 be elements of $\mathcal{P}$. Let x_1 be an element of R. Then consNE applied to x, x_1 and x_2 is an element of $\mathcal{P}$.

Definition. Let a_1 be an element of R. Let x be an element of $\mathcal{P}$. Then sdiv applied to a_1 and x is an element of SigmaType applied to $\mathcal{P}$ and the function that maps q to R.

Sdiv_rem. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$. Then $\pi_2(\mathrm{sdiv}(\mathrm{Rg}, a_1, f)) = \mathrm{ev}(\mathrm{Rg}, f, a_1)$.

SdivLen. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$ such that $|\pi_1(\mathrm{sdiv}(\mathrm{Rg}, a_1, f))| = \mathrm{pred}(|f|)$. Then $|\mathrm{consNE}(\mathrm{Rg}, f, \pi_2(\mathrm{sdiv}(\mathrm{Rg}, a_1, f)), \pi_1(\mathrm{sdiv}(\mathrm{Rg}, a_1, f)))| = |f|$.

Sdiv_len. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$. Then $|\pi_1(\mathrm{sdiv}(\mathrm{Rg}, a_1, f))| = \mathrm{pred}(|f|)$.

Definition. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$. Then rhs applied to a_1 and f is an element of $\mathcal{P}$.

FactorS. Let a_1 and c be elements of R. Let f be an element of $\mathcal{P}$ such that $f \sim \mathrm{rhs}(\mathrm{Rg}, a_1, f)$. Then $c + Xf \sim \mathrm{rhs}(\mathrm{Rg}, a_1, c + Xf)$.

Factor. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$. Then $f \sim \mathrm{rhs}(\mathrm{Rg}, a_1, f)$.

Root_factor. Let a_1 be an element of R. Let f be an element of $\mathcal{P}$ such that $\mathrm{ev}(\mathrm{Rg}, f, a_1) = 0$. Then $f \sim \mathrm{mulXa}(\mathrm{Rg}, a_1, \pi_1(\mathrm{sdiv}(\mathrm{Rg}, a_1, f)))$.

Definition. Let c and x_1 be elements of R. Let x be an element of $\mathcal{P}$. Then lastH applied to c, x and x_1 is an element of R.

Definition. Let x be an element of $\mathcal{P}$. Then lastD applied to x is an element of R.

Definition. Let c be an element of R. Let x and x_1 be elements of $\mathcal{P}$. Then initH applied to c, x and x_1 is an element of $\mathcal{P}$.

Definition. Let x be an element of $\mathcal{P}$. Then initL applied to x is an element of $\mathcal{P}$.

InitLenS. Let c be an element of R. Let f be an element of $\mathcal{P}$ such that $|\mathrm{initL}(\mathrm{Rg}, f)| = \mathrm{pred}(|f|)$. Then $|\mathrm{initH}(\mathrm{Rg}, c, f, \mathrm{initL}(\mathrm{Rg}, f))| = |f|$.

Init_len. Let f be an element of $\mathcal{P}$. Then $|\mathrm{initL}(\mathrm{Rg}, f)| = \mathrm{pred}(|f|)$.

InitPeqS. Let c be an element of R. Let f be an element of $\mathcal{P}$ such that $\mathrm{lastD}(\mathrm{Rg}, f) = 0$ implies $f \sim \mathrm{initL}(\mathrm{Rg}, f)$ and $\mathrm{lastH}(\mathrm{Rg}, c, f, \mathrm{lastD}(\mathrm{Rg}, f)) = 0$. Then $c + Xf \sim \mathrm{initH}(\mathrm{Rg}, c, f, \mathrm{initL}(\mathrm{Rg}, f))$.

Init_peq. Let f be an element of $\mathcal{P}$ such that $\mathrm{lastD}(\mathrm{Rg}, f) = 0$. Then $f \sim \mathrm{initL}(\mathrm{Rg}, f)$.

SingleS. Let c be an element of R. Let f be an element of $\mathcal{P}$ such that $|f| + 1 = 0 + 1$. Then $c + Xf \sim \mathrm{lastH}(\mathrm{Rg}, c, f, \mathrm{lastD}(\mathrm{Rg}, f)) + X0$.

Single. Let f be an element of $\mathcal{P}$ such that $|f| = 0 + 1$. Then $f \sim \mathrm{lastD}(\mathrm{Rg}, f) + X0$.

Len0. Let f be an element of $\mathcal{P}$ such that $|f| = 0$. Then $f \sim 0$.

7.3 Mul.MulR

In this section: Let Rg be a commutative ring.

Definition. Let Rg be a commutative ring. Let x and x_1 be elements of $\mathcal{P}$. Then

$x + x_1$ is an element of $\mathcal{P}$.

Coef_addP. Let p and q be elements of $\mathcal{P}$. Let k be a natural number. Then $(p+q)_k = (p_k) + (q_k)$.

Definition. Let Rg be a commutative ring. Let b be an element of R. Let x be an element of $\mathcal{P}$. Then bx is an element of $\mathcal{P}$.

Coef_scal. Let b be an element of R. Let p be an element of $\mathcal{P}$. Let k be a natural number. Then $(bp)_k = b(p_k)$.

Definition. Let Rg be a commutative ring. Let f and x be elements of $\mathcal{P}$. Then fx is an element of $\mathcal{P}$.

Coef0_pmul. Let f and g be elements of $\mathcal{P}$. Then $(fg)_0 = (f_0)(g_0)$.

Consz_peq. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Then $0+Xp \sim 0+Xq$.

Cz. Let k be a natural number. Then $(0 + X0)_k = 0$.

PinitS. Let f and g be elements of $\mathcal{P}$. Let b be an element of R. Assume that $\mathrm{lastD}(\mathrm{Rg}, g) = 0$ implies $fg \sim f\mathrm{initL}(\mathrm{Rg}, g)$. Assume that $\mathrm{lastH}(\mathrm{Rg}, b, g, \mathrm{lastD}(\mathrm{Rg}, g)) = 0$. Then $f(b + Xg) \sim f\mathrm{initH}(\mathrm{Rg}, b, g, \mathrm{initL}(\mathrm{Rg}, g))$.

Pmul_init. Let f and g be elements of $\mathcal{P}$ such that $\mathrm{lastD}(\mathrm{Rg}, g) = 0$. Then $fg \sim f\mathrm{initL}(\mathrm{Rg}, g)$.

Coef_beyond. Let f be an element of $\mathcal{P}$. Let j be a natural number. Then $f_{(|f|+j)} = 0$.

TopS. Let c be an element of R. Let f be an element of $\mathcal{P}$ such that for all natural numbers N, if $|f| = N + 1$, then $f_N = \mathrm{lastD}(\mathrm{Rg}, f)$. Let N be a natural number such that $|f| + 1$ is equal to $N + 1$. Then $(c + Xf)_N = \mathrm{lastH}(\mathrm{Rg}, c, f, \mathrm{lastD}(\mathrm{Rg}, f))$.

Coef_top. Let f be an element of $\mathcal{P}$. Let N be a natural number such that $|f| = N + 1$. Then $f_N = \mathrm{lastD}(\mathrm{Rg}, f)$.

8 Theory

Definition. Theory_fe is an element of FunExt.

Definition. propUA is an element of PropUnivalence.

9 Truncation

Definition. top_small is an element of LocallySmall applied to Base_Top.

9.1 Truncation.Trunc

In this section: Let A be a set.

Definition. Trunc_image is an element of Image applied to A, Base_Top and the function that maps a to Base_Top_tt.

Definition. An element of Trunc_carrier is an element of Image_carrier applied to A, Base_Top, the function that maps a to Base_Top_tt and Trunc_image applied to A.

Definition. Let x be an element of A. Then Trunc_inc applied to x is an element of Trunc_carrier applied to A.

Proposition. Trunc_carrier applied to A is a proposition.

Definition. Let F be a function from elements x of Trunc_carrier applied to A to sets such that for all elements x of Trunc_carrier applied to A, F applied to x is a proposition. Let x_1 be a function from elements a of A to elements of F applied to Trunc_inc applied to A and a. Let x_2 be an element of Trunc_carrier applied to A. Then Trunc_induction applied to x, F, x, a, x_1 and x_2 is an element of F applied to x_2.

Trunc_beta. Let F be a function from elements x of Trunc_carrier applied to A to sets such that for all elements x of Trunc_carrier applied to A, F applied to x is a proposition. Let b be a function from elements a of A to elements of F applied to Trunc_inc applied to A and a. Let a be an element of A. Then $\mathrm{Trunc_induction}(A, F, p, b, \mathrm{Trunc_inc}(A, a)) = b(a)$.

Uniqueness. Let F be a function from elements x of Trunc_carrier applied to A to sets such that for all elements x of Trunc_carrier applied to A, F applied to x is a proposition. Let s be a function from elements x of Trunc_carrier applied to A to elements of F applied to x. Then $\lambda k.\mathrm{Trunc_induction}(A, F, p, \lambda a.s(\mathrm{Trunc_inc}(A, a)), k) = s$.

10 Quotient

Definition. A proposition is an element of exists applied to the set of sets and the function that maps X to the proposition that X is a proposition.

Prop_path. Assume that there is a proposition. Assume that there is a propo-

sition. Let u be a function from elements x of $\pi_1(P)$ to elements of $\pi_1(Q)$. Let v be a function from elements x of $\pi_1(Q)$ to elements of $\pi_1(P)$. Then $P = Q$.

10.1 Quotient.PropFamilies

In this section: Let A be a set such that for all elements x of A, there is a proposition.

Definition. Assume that for all elements x of A, there is a proposition. Then an element of PropFamilies_Logical applied to g is an element of the function that maps g to PiType applied to A and the function that maps a to SigmaType applied to PiType applied to $\pi_1(f(a))$ and the function that maps x to $\pi_1(g(a))$ and the function that maps x to PiType applied to $\pi_1(g(a))$ and the function that maps x_1 to $\pi_1(f(a))$.

PropFamilies_logical_prop. Assume that for all elements x of A, there is a proposition. Then PropFamilies_Logical applied to A, f and g is a proposition.

Definition. logical_refl is an element of PropFamilies_Logical applied to A, f and f.

Decode. Assume that for all elements x of A, there is a proposition. Let h be an element of PropFamilies_Logical applied to A, f and g. Then $f = g$.

Definition. Assume that for all elements x of A, there is a proposition. Assume that $f = g$. Then encode applied to g and p is an element of PropFamilies_Logical applied to A, f and g.

Path_prop. Assume that for all elements x of A, there is a proposition. Then $f = g$ is a proposition.

Definition. Assume that for all elements x of A, there is a proposition. Then PropFamilies_small applied to g is an element of Small applied to $f = g$.

10.2 Quotient.Quot

In this section: Let A be a set. Let rel be a function from elements x of A and elements x_1 of A to sets such that for all elements a and b of A, rel applied to a and b is a proposition. Let rr be a function from elements a of A to elements of rel applied to a and a. Let rs be a function from elements a of A and elements b of A and elements x of rel applied to a and b to elements of rel applied to b and a. Let rt be a function from elements a of A and elements b of A and elements c of A and elements x of rel applied to a and b and elements x_1 of rel applied to b and c to elements of rel applied to a and c.

Definition. Quot_small is an element of LocallySmall applied to the proposition

that for all elements x of A, there is a proposition.

Family. Let a and x be elements of A. Then there is a proposition.

Definition. Quot_image is an element of Image applied to A, the proposition that for all elements x of A, there is a proposition and the function that maps k and k_1 to family applied to A, rel, prop, rr, rs, rt, k and k_1.

Definition. An element of Q is an element of Image_carrier applied to A, the proposition that for all elements x of A, there is a proposition, the function that maps k and k_1 to family applied to A, rel, prop, rr, rs, rt, k and k_1 and Quot_image applied to A, rel, prop, rr, rs and rt.

Definition. Let x be an element of A. Then Quot_inc applied to x is an element of Q applied to A, rel, prop, rr, rs and rt.

Definition. Let P be a function from elements x of Q applied to A, rel, prop, rr, rs and rt to sets such that for all elements q of Q applied to A, rel, prop, rr, rs and rt, P applied to q is a proposition. Let x_1 be a function from elements a of A to elements of P applied to Quot_inc applied to A, rel, prop, rr, rs, rt and a. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Then Quot_induction applied to x, P, x, a, x_1 and q is an element of P applied to q.

Code_prop. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Let a be an element of A. Then code applied to A, rel, prop, rr, rs, rt, q and a is a proposition.

Definition. Let q and r be elements of Q applied to A, rel, prop, rr, rs and rt. Then an element of Quot_Logical applied to q and r is an element of the function that maps q and r to PropFamilies_Logical applied to A, Image_inclusion applied to A, the proposition that for all elements x of A, there is a proposition, the function that maps k and k_1 to family applied to A, rel, prop, rr, rs, rt, k and k_1, Quot_image applied to A, rel, prop, rr, rs and rt and q and Image_inclusion applied to A, the proposition that for all elements x of A, there is a proposition, the function that maps k and k_1 to family applied to A, rel, prop, rr, rs, rt, k and k_1, Quot_image applied to A, rel, prop, rr, rs and rt and r.

Quot_logical_prop. Let q and r be elements of Q applied to A, rel, prop, rr, rs and rt. Then Quot_Logical applied to A, rel, prop, rr, rs, rt, q and r is a proposition.

Logical_path. Let q and r be elements of Q applied to A, rel, prop, rr, rs and rt. Let h be an element of Quot_Logical applied to A, rel, prop, rr, rs, rt, q and r. Then $q = r$.

Quot_set. Q applied to A, rel, prop, rr, rs and rt is a set.

Relates. Let a and b be elements of A. Let h be an element of rel applied to a and b. Then $\mathrm{Quot_inc}(A, \mathrm{rel}, \mathrm{prop}, \mathrm{rr}, \mathrm{rs}, \mathrm{rt}, a) = \mathrm{Quot_inc}(A, \mathrm{rel}, \mathrm{prop}, \mathrm{rr}, \mathrm{rs}, \mathrm{rt}, b)$.

Definition. Let a and b be elements of A such that Quot_inc$(A, \mathrm{rel}, \mathrm{prop}, \mathrm{rr}, \mathrm{rs}, \mathrm{rt}, a) =$ Quot_inc$(A, \mathrm{rel}, \mathrm{prop}, \mathrm{rr}, \mathrm{rs}, \mathrm{rt}, b)$. Then effective applied to a, b and p is an element of rel applied to a and b.

10.3 Quotient.Quot.Descend

In this section: Let B be a set such that B is a set. Let f be a function from elements x of A to elements of B such that for all elements a and b of A, for all elements x of rel applied to a and b, $f(a) = f(b)$.

Definition. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Let b be an element of B. Then Witness holds for q and b, if there exists an element a of A, such that Quot_inc$(A, \mathrm{rel}, \mathrm{prop}, \mathrm{rr}, \mathrm{rs}, \mathrm{rt}, a) = q$ and $f(a) = b$.

Definition. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Let b be an element of B. Then an element of Has applied to q and b is an element of the function that maps q and b to Trunc_carrier applied to Witness applied to A, rel, prop, rr, rs, rt, B, sB, f, respect, q and b.

Definition. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Then an element of Candidate applied to q is an element of the function that maps q to SigmaType applied to B and the function that maps b to Has applied to A, rel, prop, rr, rs, rt, B, sB, f, respect, q and b.

Value_eq. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Let b and c be elements of B such that Witness holds for A, rel, prop, rr, rs, rt, B, sB, f, respect, q and b and Witness holds for A, rel, prop, rr, rs, rt, B, sB, f, respect, q and c. Then $b = c$.

Unique_value. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Let b and c be elements of B. Let u be an element of Has applied to A, rel, prop, rr, rs, rt, B, sB, f, respect, q and b. Let v be an element of Has applied to A, rel, prop, rr, rs, rt, B, sB, f, respect, q and c. Then $b = c$.

Candidate_prop. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Then Candidate applied to A, rel, prop, rr, rs, rt, B, sB, f, respect and q is a proposition.

Definition. Let a be an element of A. Then at applied to a is an element of Candidate applied to A, rel, prop, rr, rs, rt, B, sB, f, respect and Quot_inc applied to A, rel, prop, rr, rs, rt and a.

Definition. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Then choose applied to q is an element of Candidate applied to A, rel, prop, rr, rs, rt, B, sB, f, respect and q.

Definition. Let q be an element of Q applied to A, rel, prop, rr, rs and rt. Then

map applied to q is an element of B.

Descend_beta. Let a be an element of A. Then map(A, rel, prop, rr, rs, rt, B, sB, f, respect, Quot_inc(A, rel, pr
$f(a)$.

10.4 RawLaws.Laws

In this section: Let A be a commutative ring.

Rr. Let p be an element of $\mathcal{P}$. Then $p \sim p$.

Laws_eq. Let p, q and r be elements of $\mathcal{P}$ such that $p \sim q$ and $q \sim r$. Then $p \sim r$.

Definition. Let p be an element of $\mathcal{P}$. Then shift applied to p is an element of $\mathcal{P}$.

Definition. Let a be an element of R. Then constantP applied to a is an element of $\mathcal{P}$.

Shift_cong. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Then shift$(A, p) \sim$ shift(A, q).

Add_cong. Let p, p$'$, q and q$'$ be elements of $\mathcal{P}$ such that $p \sim$ p$'$ and $q \sim$ q$'$. Then $p + q \sim$ p$'$ + q$'$.

Scal_cong. Let a be an element of R. Let p and q be elements of $\mathcal{P}$ such that $p \sim q$. Then $ap \sim aq$.

Scalar_cong. Let a and b be elements of R such that $a = b$. Let p be an element of $\mathcal{P}$. Then $ap \sim bp$.

Laws_add_assoc. Let p, q and r be elements of $\mathcal{P}$. Then $p+(q+r) \sim (p+q)+r$.

Laws_add_comm. Let p and q be elements of $\mathcal{P}$. Then $p + q \sim q + p$.

Laws_add_zero. Let p be an element of $\mathcal{P}$. Then $p + 0 \sim p$.

AddP_swap. Let p, q, r and s be elements of $\mathcal{P}$. Then $(p + q) + (r + s) \sim (p + r) + (q + s)$.

Scal_add. Let a be an element of R. Let p and q be elements of $\mathcal{P}$. Then $a(p + q) \sim (ap) + (aq)$.

Scal_scal. Let a and b be elements of R. Let p be an element of $\mathcal{P}$. Then $a(bp) \sim (ab)p$.

Scal_swap. Let a and b be elements of R. Let p be an element of $\mathcal{P}$. Then $a(bp) \sim b(ap)$.

Scal_one. Let p be an element of $\mathcal{P}$. Then $1p \sim p$.

Scal_zero. Let p be an element of $\mathcal{P}$. Then $0p \sim 0$.

Scal_shift. Let a be an element of R. Let p be an element of $\mathcal{P}$. Then $a\text{shift}(A, p) \sim \text{shift}(A, ap)$.

Shift_add. Let p and q be elements of $\mathcal{P}$. Then $\text{shift}(A, p + q) \sim \text{shift}(A, p) + \text{shift}(A, q)$.

Single_shift. Let a be an element of R. Let p be an element of $\mathcal{P}$. Then $\text{constantP}(A, a) + \text{shift}(A, p) \sim a + Xp$.

Scal_single. Let a and b be elements of R. Then $a\text{constantP}(A, b) \sim \text{constantP}(A, ab)$.

Definition. Let p be an element of $\mathcal{P}$. Then negP applied to p is an element of $\mathcal{P}$.

Neg_coef. Let p be an element of $\mathcal{P}$. Let k be a natural number. Then $\text{negP}(A, p)_k = -(p_k)$.

Laws_add_neg. Let p be an element of $\mathcal{P}$. Then $p + \text{negP}(A, p) \sim 0$.

10.5 RawMultiplication.Laws

In this section: Let A be a commutative ring.

Cons_cong. Let a and b be elements of R. Let p and q be elements of $\mathcal{P}$. Assume that $a = b$. Assume that $p \sim q$. Then $a + Xp \sim b + Xq$.

Mul_cong_left. Let p, p′ and q be elements of $\mathcal{P}$ such that $p \sim \text{p}'$. Then $pq \sim \text{p}'q$.

Mul_add_left. Let p, q and r be elements of $\mathcal{P}$. Then $(p + q)r \sim (pr) + (qr)$.

Mul_scal_left. Let a be an element of R. Let p and q be elements of $\mathcal{P}$. Then $(ap)q \sim a(pq)$.

Mul_shift_left. Let p and q be elements of $\mathcal{P}$. Then $\text{shift}(A, p)q \sim \text{shift}(A, pq)$.

Mul_single_left. Let a be an element of R. Let q be an element of $\mathcal{P}$. Then $\text{constantP}(A, a)q \sim aq$.

Mul_cons_left. Let a be an element of R. Let p and q be elements of $\mathcal{P}$. Then $(a + Xp)q \sim (aq) + \text{shift}(A, pq)$.

Mul_zero_left. Let q be an element of $\mathcal{P}$. Then $0q \sim 0$.

Laws_mul_comm. Let p and q be elements of $\mathcal{P}$. Then $pq \sim qp$.

Mul_cong. Let p, p′, q and q′ be elements of $\mathcal{P}$ such that $p \sim \text{p}'$ and $q \sim \text{q}'$.

Then $pq \sim \mathrm{p'q'}$.

Mul_add_right. Let p, q and r be elements of $\mathcal{P}$. Then $p(q+r) \sim (pq) + (pr)$.

Mul_scal_right. Let a be an element of R. Let p and q be elements of $\mathcal{P}$. Then $p(aq) \sim a(pq)$.

Mul_shift_right. Let p and q be elements of $\mathcal{P}$. Then $p\mathrm{shift}(A, q) \sim \mathrm{shift}(A, pq)$.

Laws_mul_assoc. Let p, q and r be elements of $\mathcal{P}$. Then $p(qr) \sim (pq)r$.

Laws_mul_one. Let p be an element of $\mathcal{P}$. Then $p\mathrm{constantP}(A, 1) \sim p$.

11 RingHom

Definition. Let A and B be commutative rings. Let f be a function from elements x of R to elements of R. Then AddLaw holds for A, B, x and f, if for all elements a and b of R, $x(a+b) = x(a) + x(b)$.

Definition. Let A and B be commutative rings. Let f be a function from elements x of R to elements of R. Then MulLaw holds for A, B, x and f, if for all elements a and b of R, $x(ab) = x(a)x(b)$.

Definition. Let A and B be commutative rings. Let f be a function from elements x of R to elements of R. Then Preserves holds for A, B, x and f, if $x(1) = 1$, for all elements a of R, for all elements b of R, $x(a+b) = x(a) + x(b)$ and for all elements a of R, for all elements b of R, $x(ab) = x(a)x(b)$.

Definition. Let A and B be commutative rings. Then Hom holds for A and B, if there exists an element f of PiType applied to R and the function that maps x to R, such that Preserves holds for A, B and f.

11.1 RingHom.Properties

In this section: Let A and B be commutative rings. Assume that R is a set.

Prop. Let f be a function from elements x of R to elements of R. Then Preserves applied to A, B and f is a proposition.

Properties_set. Hom applied to A and B is a set.

Ext. Assume that Hom holds for A and B. Assume that Hom holds for A and B. Assume that for all elements a of R, $\pi_1(f)a = \pi_1(g)a$. Then $f = g$.

12 RingHom

Identity. Let A be a commutative ring. Then Hom holds for A and A.

Compose. Let A, B and C be commutative rings such that Hom holds for B and C and Hom holds for A and B. Then Hom holds for A and C.

RingHom_zero. Let A and B be commutative rings such that Hom holds for A and B. Then $\pi_1(f)0 = 0$.

Negation. Let A and B be commutative rings such that Hom holds for A and B. Let a be an element of R. Then $\pi_1(f)-a = -(\pi_1(f)a)$.

12.1 PolynomialRing.Extension

In this section: Let A be a commutative ring. Assume that R is a set.

Peq_prop. Let p and q be elements of $\mathcal{P}$. Then $p \sim q$ is a proposition.

Definition. Let A be a commutative ring. Assume that R is a set. Then $A[X]$ is a set defined as Q applied to $\mathcal{P}$, the function that maps k and k_1 to $k \sim k_1$, the function that maps k and k_1 to peq_prop applied to A, setA, k and k_1, the function that maps k to rr applied to A and k, the function that maps k, k_1 and k_2 to peq_sym applied to A, k, k_1 and k_2 and the function that maps k, k_1, k_2, k_3 and k_4 to peq_trans applied to A, k, k_1, k_2, k_3 and k_4.

Definition. Let A be a commutative ring. Assume that R is a set. Let x be an element of $\mathcal{P}$. Then $[x]$ is an element of $A[X]$.

Extension_set. $A[X]$ is a set.

12.2 PolynomialRing.Extension.Binary

In this section: Let op be a function from elements x of $\mathcal{P}$ and elements x_1 of $\mathcal{P}$ to elements of $\mathcal{P}$ such that for all elements p, p′, q and q′ of $\mathcal{P}$, if $p \sim$ p′ and $q \sim$ q′, then $\mathrm{op}(p, q) \sim \mathrm{op}(\mathrm{p}', \mathrm{q}')$.

Left_respects. Let p and p′ be elements of $\mathcal{P}$ such that $p \sim$ p′. Then λk′.map($\mathcal{P}$, $\lambda k.\lambda k_1.k \sim$
$k_1, \lambda k.\lambda k_1.\mathrm{peq_prop}(A, \mathrm{setA}, k, k_1), \lambda k.\mathrm{rr}(A, k), \lambda k.\lambda k_1.\lambda k_2.\mathrm{peq_sym}(A, k, k_1, k_2), \lambda k.\lambda k_1.\lambda k_2.\lambda k_3.\lambda k_4.\mathrm{peq_tr}$
$k_1, \lambda k.\lambda k_1.\mathrm{peq_prop}(A, \mathrm{setA}, k, k_1), \lambda k.\mathrm{rr}(A, k), \lambda k.\lambda k_1.\lambda k_2.\mathrm{peq_sym}(A, k, k_1, k_2), \lambda k.\lambda k_1.\lambda k_2.\lambda k_3.\lambda k_4.\mathrm{peq_tr}$
λk′.map($\mathcal{P}$, $\lambda k.\lambda k_1.k \sim k_1, \lambda k.\lambda k_1.\mathrm{peq_prop}(A, \mathrm{setA}, k, k_1), \lambda k.\mathrm{rr}(A, k), \lambda k.\lambda k_1.\lambda k_2.\mathrm{peq_sym}(A, k, k_1, k_2), \lambda k$
$k_1, \lambda k.\lambda k_1.\mathrm{peq_prop}(A, \mathrm{setA}, k, k_1), \lambda k.\mathrm{rr}(A, k), \lambda k.\lambda k_1.\lambda k_2.\mathrm{peq_sym}(A, k, k_1, k_2), \lambda k.\lambda k_1.\lambda k_2.\lambda k_3.\lambda k_4.\mathrm{peq_tr}$

Definition. Let x and x_1 be elements of $A[X]$. Then operation applied to x and x_1 is an element of $A[X]$.

Binary_beta. Let p and q be elements of $\mathcal{P}$. Then operation$(A, \mathrm{setA}, \mathrm{op}, \mathrm{cong}, [p], [q]) = [\mathrm{op}(p, q)]$.

12.3 PolynomialRing.Extension

In this section: Let A be a commutative ring. Assume that R is a set.

Definition. Let A be a commutative ring. Assume that R is a set. Let x and x_1 be elements of $A[X]$. Then $x + x_1$ is an element of $A[X]$.

Definition. Let A be a commutative ring. Assume that R is a set. Let x and x_1 be elements of $A[X]$. Then xx_1 is an element of $A[X]$.

Definition. Let A be a commutative ring. Assume that R is a set. Let x be an element of $A[X]$. Then $-x$ is an element of $A[X]$.

Definition. Let A be a commutative ring. Assume that R is a set. Then 0 is an element of $A[X]$.

Definition. Let A be a commutative ring. Assume that R is a set. Then 1 is an element of $A[X]$.

Definition. Let B be a function from elements x of $A[X]$ and elements x_1 of $A[X]$ to sets such that for all elements x and y of $A[X]$, B applied to x and y is a proposition. Let b be a function from elements p of $\mathcal{P}$ and elements q of $\mathcal{P}$ to elements of B applied to $[p]$ and $[q]$. Let x and y be elements of $A[X]$. Then forall2 applied to x, x_1, B, prop, p, q, b, x and y is an element of B applied to x and y.

Definition. Let B be a function from elements x of $A[X]$ and elements x_1 of $A[X]$ and elements x_2 of $A[X]$ to sets such that for all elements x, y and z of $A[X]$, B applied to x, y and z is a proposition. Let b be a function from elements p of $\mathcal{P}$ and elements q of $\mathcal{P}$ and elements r of $\mathcal{P}$ to elements of B applied to $[p]$, $[q]$ and $[r]$. Let x, y and z be elements of $A[X]$. Then forall3 applied to x, x_1, x_2, B, prop, p, q, r, b, x, y and z is an element of B applied to x, y and z.

Definition. The elements of Expr are a basic type.

Definition. Let p be an element of $\mathcal{P}$. Then atom applied to p is an element of Expr applied to A and setA.

Definition. Let a and b be elements of Expr applied to A and setA. Then sumE applied to a and b is an element of Expr applied to A and setA.

Definition. Let a and b be elements of Expr applied to A and setA. Then mulE applied to a and b is an element of Expr applied to A and setA.

Definition. Let a be an element of Expr applied to A and setA. Then negE

applied to a is an element of Expr applied to A and setA.

Definition. Let x be an element of Expr applied to A and setA. Then raw applied to x is an element of $\mathcal{P}$.

Definition. Let x be an element of Expr applied to A and setA. Then value applied to x is an element of $A[X]$.

Sound. Let e be an element of Expr applied to A and setA. Then $\text{value}(A, \text{setA}, e) = [\text{raw}(A, \text{setA}, e)]$.

Extension_lift. Let e and e' be elements of Expr applied to A and setA such that $\text{raw}(A, \text{setA}, e) \sim \text{raw}(A, \text{setA}, \text{e}')$. Then $\text{value}(A, \text{setA}, e) = \text{value}(A, \text{setA}, \text{e}')$.

Extension_add_assoc. Let x, y and z be elements of $A[X]$. Then $x + (y + z) = (x + y) + z$.

Extension_add_comm. Let x and y be elements of $A[X]$. Then $x + y = y + x$.

Extension_add_zero. Let x be an element of $A[X]$. Then $x + 0 = x$.

Extension_add_neg. Let x be an element of $A[X]$. Then $x + -x = 0$.

Extension_mul_assoc. Let x, y and z be elements of $A[X]$. Then $x(yz) = (xy)z$.

Extension_mul_comm. Let x and y be elements of $A[X]$. Then $xy = yx$.

Extension_mul_one. Let x be an element of $A[X]$. Then $x1 = x$.

Distributive. Let x, y and z be elements of $A[X]$. Then $x(y + z) = (xy) + (xz)$.

Definition. Let A be a commutative ring. Assume that R is a set. Then $A[X]$ is a commutative ring.

Definition. Let a be an element of R. Then constant applied to a is an element of $A[X]$.

Definition. variable is an element of $A[X]$.

Constant_add. Let a and b be elements of R. Then $\text{constant}(A, \text{setA}, a + b) = \text{constant}(A, \text{setA}, a) + \text{constant}(A, \text{setA}, b)$.

Constant_mul. Let a and b be elements of R. Then $\text{constant}(A, \text{setA}, ab) = \text{constant}(A, \text{setA}, a)\text{constant}(A, \text{setA}, b)$.

Extension_inclusion. Hom holds for A and $A[X]$.

Definition. Extension_T is an element of $\mathcal{P}$.

T_mul. Let p be an element of $\mathcal{P}$. Then $\text{Extension_T}(A, \text{setA})p \sim \text{shift}(A, p)$.

Horner. Let a be an element of R. Let p be an element of $\mathcal{P}$. Then $[(a + Xp)] = \text{constant}(A, \text{setA}, a) + (\text{variable}(A, \text{setA})[p])$.

Constant_injective. Let a and b be elements of R such that constant$(A, \text{setA}, a) =$ constant(A, setA, b). Then $a = b$.

12.4 Coefficients.Over

In this section: Let A be a commutative ring. Assume that R is a set.

Definition. An element of Coefficients is an element of PiType applied to $\mathbb{N}$ and the function that maps x to R.

Definition. Let x be an element of $A[X]$. Let k be a natural number. Then Coefficients_Over_coef applied to x and k is an element of R.

Coef_include. Let p be an element of $\mathcal{P}$. Let k be a natural number. Then Coefficients_Over_coef$(A, \text{setA}, [p], k) = p_k$.

Definition. Let lhs and rhs be functions from elements x of $A[X]$ and natural numbers x_1 to elements of R. Let x be an element of $A[X]$. Then CoefLaw holds for x, x_1, lhs, x, x_1, rhs and x, if for all elements k of $\mathbb{N}$, $x(\text{lhs}, k) = x_1(\text{lhs}, k)$.

Law. Let lhs and rhs be functions from elements x of $A[X]$ and natural numbers x_1 to elements of R such that for all elements p of $\mathcal{P}$, CoefLaw holds for A, setA, lhs, rhs and $[p]$. Let x be an element of $A[X]$. Then CoefLaw holds for A, setA, lhs, rhs and x.

Coef_ext. Let x and y be elements of $A[X]$ such that for all natural numbers k, Coefficients_Over_coef(A, setA, x, k) = Coefficients_Over_coef(A, setA, y, k). Then $x = y$.

Coef_add. Let x and y be elements of $A[X]$. Let k be a natural number. Then Coefficients_Over_coef$(A, \text{setA}, x{+}y, k)$ = Coefficients_Over_coef$(A, \text{setA}, x, k)+$ Coefficients_Over_coef(A, setA, y, k).

Coef_scale. Let a be an element of R. Let x be an element of $A[X]$. Let k be a natural number. Then Coefficients_Over_coef$(A, \text{setA}, \text{constant}(A, \text{setA}, a)x, k) =$ aCoefficients_Over_coef(A, setA, x, k).

Coef_neg. Let x be an element of $A[X]$. Let k be a natural number. Then Coefficients_Over_coef$(A, \text{setA}, -x, k) = -$Coefficients_Over_coef$(A, \text{setA}, x, k)$.

Definition. Let x be an element of $A[X]$. Let x_1 be a natural number. Then shifted applied to x and x_1 is an element of R.

Shift_coef. Let p be an element of $\mathcal{P}$. Let k be a natural number. Then $\text{shift}(A, p)_k = \text{shifted}(A, \text{setA}, [p], k)$.

Coef_variable. Let x be an element of $A[X]$. Let k be a natural number. Then Coefficients_Over_coef$(A, \text{setA}, \text{variable}(A, \text{setA})x, k) = \text{shifted}(A, \text{setA}, x, k)$.

Coef_zero. Let k be a natural number. Then Coefficients_Over_coef$(A, \mathrm{setA}, 0, k) = 0$.

Coef_const0. Let a be an element of R. Then Coefficients_Over_coef$(A, \mathrm{setA}, \mathrm{constant}(A, \mathrm{setA}, a), 0) = a$.

Coef_constS. Let a be an element of R. Let k be a natural number. Then Coefficients_Over_coef$(A, \mathrm{setA}, \mathrm{constant}(A, \mathrm{setA}, a), k + 1) = 0$.

Definition. Let d be a natural number. Let x be an element of $A[X]$. Then Coefficients_Over_Bounded holds for d and x, if for all natural numbers i, Coefficients_Over_coef$(A, \mathrm{setA}, x, d + i) = 0$.

13 NatOrder

Definition. Let x and x_1 be natural numbers. Then the elements of Le applied to x and x_1 are a basic type.

Definition. Let m and n be natural numbers. Then $m < n$, if Le holds for $m+1$ and n.

Definition. Let n be a natural number. Then le_refl applied to n is an element of Le applied to n and n.

Definition. Let a, b and c be natural numbers. Let x be an element of Le applied to a and b. Let x_1 be an element of Le applied to b and c. Then le_trans applied to a, b, c, x and x_1 is an element of Le applied to a and c.

Definition. Let m and n be natural numbers. Then le_or_gt applied to m and n is an element of Plus applied to Le applied to m and n and Lt(n, m).

Le_plus. Let m and n be natural numbers. Let x be an element of Le applied to m and n. Then there exists a natural number d, such that $n = d + m$.

Definition. Let m and n be natural numbers. Let x be an element of Le applied to m and n. Then le_suc applied to m, n and x is an element of Le applied to m and $n + 1$.

Definition. Let s and d be natural numbers. Then le_plus_right applied to s and d is an element of Le applied to d and $s + d$.

Definition. Let r and k be natural numbers. Let x be an element of Plus applied to Lt(r, k) and $r = k$. Then lt_or_eq_suc applied to r, k and x is an element of Plus applied to Lt$(r + 1, k + 1)$ and $r + 1 = k + 1$.

Definition. Let r and k be natural numbers. Let x be an element of Le applied to r and k. Then le_cases applied to r, k and x is an element of Plus applied to Lt(r, k) and $r = k$.

Compare_suc. Let m and n be natural numbers such that there exists a natural number d such that m is equal to $n + d$ or there exists a natural number d such that $n = m+d$. Then there exists a natural number d such that $m+1 = n+1+d$ or there exists an element d of $\mathbb{N}$ such that $n + 1 = m + 1 + d$.

Compare. Let m and n be natural numbers. Then there exists a natural number d such that $m = n + d$ or there exists a natural number d such that $n = m + d$.

14 GcdDomain

Empty_prop. $\emptyset$ is a proposition.

Not_prop. Let X be a set. Then $\neg X$ is a proposition.

14.1 GcdDomain.Divisibility

In this section: Let A be a commutative ring.

Definition. Let A be a commutative ring. Let a and b be elements of R. Then a divides b, if there exists an element q of R, such that $b = aq$.

Div_refl. Let a be an element of R. Then a divides a.

Div_zero. Let a be an element of R. Then a divides 0.

Div_one. Let a be an element of R. Then 1 divides a.

Div_trans. Let a, b and c be elements of R such that a divides b and b divides c. Then a divides c.

Div_mul. Let a, b and c be elements of R such that a divides b. Then a divides bc.

Div_mul_left. Let a, b and c be elements of R such that a divides b. Then a divides cb.

Div_factor. Let a and c be elements of R. Then a divides ac.

Div_factor_left. Let a and c be elements of R. Then a divides ca.

Div_add. Let a, b and c be elements of R such that a divides b and a divides c. Then a divides $b + c$.

Div_neg. Let a and b be elements of R such that a divides b. Then a divides $-b$.

Div_scale. Let a, b and c be elements of R such that a divides b. Then ca divides

cb.

Div_mul2. Let a, b, c and d be elements of R such that a divides b and c divides d. Then ac divides bd.

Div_eq_left. Let a, a' and b be elements of R such that $a = \mathrm{a}'$ and a divides b. Then a' divides b.

Div_eq_right. Let a, b and b' be elements of R such that $b = \mathrm{b}'$ and a divides b. Then a divides b'.

Div_of_zero. Let b be an element of R such that 0 divides b. Then $b = 0$.

Definition. Let A be a commutative ring. Let a and b be elements of R. Then a and b have a greatest common divisor, if there exists an element d of R, such that d divides a, d divides b and for all elements c of R, if c divides a, then if c divides b, then c divides d.

15 GcdDomain

Definition. A gcd domain on A is an algebraic structure, where

- A is a commutative ring.
- R is a set.
- for all elements x of R, $x = 0$ or x is not equal to 0.
- for all elements x and y of R, if x is not equal to 0 and y is not equal to 0, then xy is not equal to 0.
- for all elements a and b of R, a and b have a greatest common divisor.
- 1 is not equal to 0.

Definition. Let A be a commutative ring. Assume that R is a set. Assume that for all elements x of R, $x = 0$ or x is not equal to 0. Assume that for all elements x and y of R, if x is not equal to 0 and y is not equal to 0, then xy is not equal to 0. Assume that for all elements a and b of R, a and b have a greatest common divisor. Assume that 1 is not equal to 0. Then mkGcdDomain applied to A, set, dec, integral, gcd and nontrivial is a gcd domain on A.

15.1 GcdDomain.Domain

In this section: Let A be a commutative ring. Assume that R is a set. Assume that for all elements x of R, $x = 0$ or x is not equal to 0. Assume that for all

elements x and y of R, if x is not equal to 0 and y is not equal to 0, then xy is not equal to 0.

Mul_sub. Let u, a and b be elements of R. Then $u(a + -b) = (ua) + -(ub)$.

Sub_zero. Let a and b be elements of R such that $a + -b = 0$. Then $a = b$.

Sub_self. Let a be an element of R. Then $a + -a = 0$.

Cancel_cancelD. Let u, a and b be elements of R such that u is not equal to 0, ua is equal to ub and either $a + -b = 0$ or $a + -b$ is not equal to 0. Then $a = b$.

Domain_cancel. Let u, a and b be elements of R such that u is not equal to 0 and $ua = ub$. Then $a = b$.

Cancel_zero. Let u and x be elements of R such that u is not equal to 0 and $ux = 0$. Then $x = 0$.

Div_cancel. Let u, a and b be elements of R such that u is not equal to 0 and ua divides ub. Then a divides b.

Div_prop. Let a and b be elements of R such that a is not equal to 0. Then $a \mid b$ is a proposition.

Nonzero_prop. Let x be an element of R. Then the proposition that x is not equal to 0 is a proposition.

Mul_nonzero. Let x and y be elements of R such that x is not equal to 0 and y is not equal to 0. Then xy is not equal to 0.

15.2 GcdDomain.Gcd

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let a and b be elements of R. Then $\gcd(a, b)$ is an element of R.

Gcd_left. Let a and b be elements of R. Then $\gcd(a, b)$ divides a.

Gcd_right. Let a and b be elements of R. Then $\gcd(a, b)$ divides b.

Gcd_univ. Let a, b and c be elements of R such that c divides a and c divides b. Then c divides $\gcd(a, b)$.

15.3 GcdDomain.Gcd.M

In this section: Let u, a, b and d be elements of R such that u is not equal to 0, d divides ua and d divides ub.

Definition. g is an element of R.

M_dg. $d \mid g(A, G, u, a, b, d, \mathrm{hu}, h_1, h_2)$.

Ug. $u \mid g(A, G, u, a, b, d, \mathrm{hu}, h_1, h_2)$.

Definition. M_t is an element of R.

Ta. $\mathrm{M_t}(A, G, u, a, b, d, \mathrm{hu}, h_1, h_2) \mid a$.

Tb. $\mathrm{M_t}(A, G, u, a, b, d, \mathrm{hu}, h_1, h_2) \mid b$.

Gu. $g(A, G, u, a, b, d, \mathrm{hu}, h_1, h_2) \mid u(\gcd(a, b))$.

M_result. d divides $u(\gcd(a, b))$.

15.4 GcdDomain.Gcd

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Lemma_M_byU. Let u, a, b and d be elements of R such that d divides ua, d divides ub and either u is equal to 0 or u is not equal to 0. Then d divides $u(\gcd(a, b))$.

Lemma_M. Let u, a, b and d be elements of R such that d divides ua and d divides ub. Then d divides $u(\gcd(a, b))$.

Euclid. Let a, b and c be elements of R such that $\gcd(a, b)$ divides 1 and a divides bc. Then a divides c.

15.5 GcdDomain.FieldGcd

In this section: Let F be a commutative ring. Assume that R is a set. Assume that for all elements x of R, $x = 0$ or there exists an element y of R such that $xy = 1$. Assume that 1 is not equal to 0.

Unit_nonzero. Let x be an element of R such that there exists an element y of R such that $xy = 1$ and $x = 0$. Then we have a contradiction.

DecZ_decD. Let x be an element of R such that $x = 0$ or there exists an element y of R such that $xy = 1$. Then $x = 0$ or x is not equal to 0.

DecZ. Let x be an element of R. Then $x = 0$ or x is not equal to 0.

Integral_intD. Let x and y be elements of R such that x is not equal to 0, y is not equal to 0 and either $x = 0$ or there exists an element w of R such that xw is equal to 1. Then xy is not equal to 0.

FieldGcd_integral. Let x and y be elements of R such that x is not equal to 0 and y is not equal to 0. Then xy is not equal to 0.

Unit_div. Let x and b be elements of R such that there exists an element y of R, such that $xy = 1$. Then x divides b.

Gcd_gcdD. Let a and b be elements of R such that $a = 0$ or there exists an element y of R such that $ay = 1$. Then a and b have a greatest common divisor.

FieldGcd_gcd. Let a and b be elements of R. Then a and b have a greatest common divisor.

Definition. FieldGcd_domain is a gcd domain on F.

15.6 Content.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let n be a natural number. Let P be a function from natural numbers x to sets. Then an element of Below applied to n, x and P is an element of the function that maps n, P and P to PiType applied to $\mathbb{N}$, the function that maps i to PiType applied to $\mathrm{Lt}(i, n)$ and the function that maps x_1 to x applied to i and P.

Definition. Let n be a natural number. Let e be a function from natural numbers x to elements of R. Then Bnd holds for n, x and e, if for all natural numbers i, $x(n + i) = 0$.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let e be a function from natural numbers x to elements of R. Then x is primitive, if for all elements d of R, if for all natural numbers i, $d \mid x(i)$, then d divides 1.

Definition. Let n be a natural number. Then lt_top applied to n is an element of $\mathrm{Lt}(n, n + 1)$.

Definition. Let i and n be natural numbers. Let h be an element of $\mathrm{Lt}(i, n)$. Then lt_up applied to i, n and h is an element of $\mathrm{Lt}(i, n + 1)$.

Definition. Let e be a function from natural numbers x to elements of R. Let x be a natural number. Then cont applied to x, e and x is an element of R.

Cont_div_step. Let e be a function from natural numbers x to elements of R. Let n and i be natural numbers. Let r be an element of Below applied to A, G, n and the function that maps i to $\mathrm{cont}(A, G, e, n) \mid e(i)$. Let x be an element of Plus applied to $\mathrm{Lt}(i, n)$ and $i = n$. Then $\mathrm{cont}(A, G, e, n + 1) \mid e(i)$.

Definition. Let e be a function from natural numbers x to elements of R. Let n be a natural number. Then cont_div applied to x, e and n is an element of

Below applied to A, G, n and the function that maps i to $\mathrm{cont}(A, G, e, n) \mid e(i)$.

Cont_univ. Let e be a function from natural numbers x to elements of R. Let d be an element of R. Let n be a natural number. Let x be an element of Below applied to A, G, n and the function that maps i to $d \mid e(i)$. Then $d \mid \mathrm{cont}(A, G, e, n)$.

Cont_M. Let e be a function from natural numbers x to elements of R. Let u and d be elements of R. Let n be a natural number. Let x be an element of Below applied to A, G, n and the function that maps i to $d \mid (ue(i))$. Then d divides $u\mathrm{cont}(A, G, e, n)$.

15.7 Content.Over.Bounded

In this section: Let e be a function from natural numbers x to elements of R. Let n be a natural number such that Bnd holds for A, G, n and e.

Every_step. Let d be an element of R. Let h be an element of Below applied to A, G, n and the function that maps i to $d \mid e(i)$. Let i be a natural number. Let x be an element of Plus applied to Le applied to n and i and $\mathrm{Lt}(i, n)$. Then $d \mid e(i)$.

Every. Let d be an element of R. Let h be an element of Below applied to A, G, n and the function that maps i to $d \mid e(i)$. Let i be a natural number. Then $d \mid e(i)$.

Definition. Bounded_c is an element of R.

C_div. Let i be a natural number. Then $\mathrm{Bounded_c}(A, G, e, n, \mathrm{bnd}) \mid e(i)$.

C_univ. Let d be an element of R such that for all natural numbers i, $d \mid e(i)$. Then $d \mid \mathrm{Bounded_c}(A, G, e, n, \mathrm{bnd})$.

C_zero. Assume that $\mathrm{Bounded_c}(A, G, e, n, \mathrm{bnd}) = 0$. Let i be a natural number. Then $e(i) = 0$.

Prim_c. Assume that e is primitive. Then $\mathrm{Bounded_c}(A, G, e, n, \mathrm{bnd}) \mid 1$.

Prim_M. Assume that e is primitive. Let u and d be elements of R such that for all natural numbers i, d divides $ue(i)$. Then d divides u.

15.8 PolyBase.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Then $A[X]$ is a set defined as $A[X]$.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let x be an element of $A[X]$. Let k be a natural number. Then x_k is an element of R.

Definition. Let a be an element of R. Let x be an element of $A[X]$. Then sc applied to a and x is an element of $A[X]$.

Coef_sc. Let a be an element of R. Let x be an element of $A[X]$. Let k be a natural number. Then $\mathrm{sc}(A, G, a, x)_k = a(x_k)$.

Coef_mul_include. Let p and q be elements of $\mathcal{P}$. Let k be a natural number. Then $([p][q])_k = (pq)_k$.

Definition. Let x be an element of $A[X]$. Then PolyBase_Over_Prim holds for x, if the function that maps k to x_k is primitive.

Prim_include. Let p be an element of $\mathcal{P}$ such that PolyBase_Over_Prim holds for A, G and $[p]$. Then the function that maps k to p_k is primitive.

Bnd_list. Let p be an element of $\mathcal{P}$. Then Bnd holds for A, G, $|p|$ and the function that maps k to p_k.

Definition. Let a be an element of R. Let q be an element of $\mathcal{P}$ such that for all natural numbers i, a divides q_i. Then divL applied to a, q and x is an element of $\mathcal{P}$.

DivL_coef. Let a be an element of R. Let q be an element of $\mathcal{P}$ such that for all natural numbers i, a divides q_i. Let i be a natural number. Then $q_i = a(\mathrm{divL}(A, G, a, q, w)_i)$.

15.9 PolyBase.Over.Scalar

In this section: Let a be an element of R such that a is not equal to 0.

Definition. Let x be an element of $A[X]$. Then Quot holds for x, if there exists an element y of $A[X]$, such that $x = \mathrm{sc}(A, G, a, y)$.

Scalar_cancel. Let y and y' be elements of $A[X]$ such that $\mathrm{sc}(A, G, a, y) = \mathrm{sc}(A, G, a, \mathrm{y}')$. Then $y = \mathrm{y}'$.

Quot_prop. Let x be an element of $A[X]$. Then Quot applied to A, G, a, ha and x is a proposition.

Definition. Let x be an element of $A[X]$. Then Divides holds for x, if for all natural numbers i, a divides x_i.

Divide_list. Let q be an element of $\mathcal{P}$ such that Divides holds for A, G, a, ha and $[q]$. Then Quot holds for A, G, a, ha and $[q]$.

Definition. Let x be an element of $A[X]$ such that Divides holds for A, G, a, ha and x. Then we can say that x divides x_1.

15.10 Gauss.Direct

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let p be an element of $\mathcal{P}$. Then $\mathrm{c}(p)$ is an element of R.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let d be an element of R. Let p be an element of $\mathcal{P}$. Then d is a common divisor of the coefficients of p, if for all natural numbers i, d divides p_i.

Definition. Let p and q be elements of $\mathcal{P}$. Then Product holds for p and q, if for all elements d of R, if d is a common divisor of the coefficients of pq, then d divides $\mathrm{c}(p)\mathrm{c}(q)$.

Div_sub. Let d, a and b be elements of R such that d divides a and d divides b. Then d divides $a + -b$.

Sub_add. Let x and y be elements of R. Then $(x + y) + -x = y$.

15.11 Gauss.Direct.Tail

In this section: Let a and d be elements of R. Let p and q be elements of $\mathcal{P}$. Assume that d divides a. Assume that d is a common divisor of the coefficients of $(a + Xp)q$. Let i be a natural number.

Definition. x is an element of R.

Definition. y is an element of R.

Tail_eq. $((a + Xp)q)_{i+1} = x(A, G, a, p, q, d, \mathrm{ha}, h, i) + y(A, G, a, p, q, d, \mathrm{ha}, h, i)$.

Dx. $d \mid x(A, G, a, p, q, d, \mathrm{ha}, h, i)$.

Dxy. d divides $x(A, G, a, p, q, d, \mathrm{ha}, h, i) + y(A, G, a, p, q, d, \mathrm{ha}, h, i)$.

Tail_result. $d \mid y(A, G, a, p, q, d, \mathrm{ha}, h, i)$.

15.12 Gauss.Direct

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Tail_common. Let a and d be elements of R. Let p and q be elements of $\mathcal{P}$.

Assume that d divides a. Assume that d is a common divisor of the coefficients of $(a + Xp)q$. Let i be a natural number. Then d divides $(pq)_i$.

Definition. Let N be a natural number. Then Direct_Rec holds for N, if for all elements p and q of $\mathcal{P}$, if $|p| + |q| = N$, then Product holds for A, G, p and q.

15.13 Gauss.Direct.PrimStep

In this section: Let N be a natural number such that Direct_Rec holds for A, G and N. Let a, b and d be elements of R. Let p and q be elements of $\mathcal{P}$ such that $|(a+Xp)|+|(b+Xq)| = N+1$, the function that maps k to $(a + Xp)_k$ and the function that maps k to $(b + Xq)_k$ are primitive and d is a common divisor of the coefficients of $(a + Xp)(b + Xq)$.

PrimStep_ef. $|p| + |(b + Xq)| = N$.

Es. $|q| + |(a + Xp)| = N$.

15.14 Gauss.Direct.PrimStep.One

In this section: Let a_0 be an element of R. Let p_0 and q_0 be elements of $\mathcal{P}$ such that $|p_0| + |q_0| = N$, the function that maps k to $(a_0 + Xp_0)_k$ and the function that maps k to q_{0_k} are primitive and d is a common divisor of the coefficients of $(a_0 + Xp_0)q_0$.

Definition. One_t is an element of R.

Ht. One_t applied to A, G, N, r, a, p, b, q, e, pp, pq, d, h, a_0, p_0, q_0, ep, prim0, prim1 and hh is a common divisor of the coefficients of $(a_0 + Xp_0)q_0$.

Tc. One_t$(A, G, N, r, a, p, b, q, e, \text{pp}, \text{pq}, d, h, a_0, p_0, q_0, \text{ep}, \text{prim0}, \text{prim1}, \text{hh}) \mid \text{c}(p_0)\text{c}(q_0)$.

Ct. One_t$(A, G, N, r, a, p, b, q, e, \text{pp}, \text{pq}, d, h, a_0, p_0, q_0, \text{ep}, \text{prim0}, \text{prim1}, \text{hh}) \mid \text{c}(p_0)$.

Coeffs. One_t applied to A, G, N, r, a, p, b, q, e, pp, pq, d, h, a_0, p_0, q_0, ep, prim0, prim1 and hh is a common divisor of the coefficients of $a_0 + Xp_0$.

One_result. One_t$(A, G, N, r, a, p, b, q, e, \text{pp}, \text{pq}, d, h, a_0, p_0, q_0, \text{ep}, \text{prim0}, \text{prim1}, \text{hh}) \mid 1$.

15.15 Gauss.Direct.PrimStep

In this section: Let N be a natural number such that Direct_Rec holds for A, G and N. Let a, b and d be elements of R. Let p and q be elements of $\mathcal{P}$ such that $|(a+Xp)|+|(b+Xq)| = N+1$, the function that maps k to $(a + Xp)_k$ and

the function that maps k to $(b + Xq)_k$ are primitive and d is a common divisor of the coefficients of $(a + Xp)(b + Xq)$.

CpA. $\gcd(d, a)$ divides 1.

Hs. d is a common divisor of the coefficients of $(b + Xq)(a + Xp)$.

CpB. $\gcd(d, b)$ divides 1.

Ab. d divides ab.

Db. d divides b.

PrimStep_result. d divides 1.

15.16 Gauss.Direct

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Not_prim_nil. Assume that the function that maps k to 0_k is primitive. Then we have a contradiction.

Primitive. Let N be a natural number such that Direct_Rec holds for A, G and N. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = N + 1$ and the function that maps k to p_k and the function that maps k to q_k are primitive. Then the function that maps k to $(pq)_k$ is primitive.

Div_len. Let c be an element of R. Let p be an element of $\mathcal{P}$ such that c is a common divisor of the coefficients of p. Then $|\mathrm{divL}(A, G, c, p, w)| = |p|$.

15.17 Gauss.Direct.Part

In this section: Let p be an element of $\mathcal{P}$ such that $\mathrm{c}(p)$ is not equal to 0.

Definition. Part_c is an element of R.

Definition. qs is an element of $\mathcal{P}$.

Definition. length is an element of Proof applied to $|\mathrm{qs}(A, G, p, \mathrm{nc})| = |p|$.

Part_eq. Let i be a natural number. Then $p_i = \mathrm{Part_c}(A, G, p, \mathrm{nc})(\mathrm{qs}(A, G, p, \mathrm{nc})_i)$.

Part_scaled. $p \sim \mathrm{Part_c}(A, G, p, \mathrm{nc})\mathrm{qs}(A, G, p, \mathrm{nc})$.

Part_prim. The function that maps k to $\mathrm{qs}(A, G, p, \mathrm{nc})_k$ is primitive.

15.18 Gauss.Direct.Nonzero

In this section: Let N be a natural number such that Direct_Rec holds for A, G and N. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = N + 1$, $\mathrm{c}(p)$ is not equal to 0 and $\mathrm{c}(q)$ is not equal to 0.

Definition. Nonzero_u is an element of R.

Len'. $|\mathrm{qs}(A, G, p, \mathrm{np})| + |\mathrm{qs}(A, G, q, \mathrm{nq})| = N + 1$.

Nonzero_prim. The function that maps k to $(\mathrm{qs}(A, G, p, \mathrm{np})\mathrm{qs}(A, G, q, \mathrm{nq}))_k$ is primitive.

Nonzero_scaled. $pq \sim \mathrm{Nonzero_u}(A, G, N, r, p, q, e, \mathrm{np}, \mathrm{nq})(\mathrm{qs}(A, G, p, \mathrm{np})\mathrm{qs}(A, G, q, \mathrm{nq}))$.

Nonzero_result. Product holds for A, G, p and q.

15.19 Gauss.Direct

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Step_byQ. Let N be a natural number such that Direct_Rec holds for A, G and N. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = N + 1$, $\mathrm{c}(p)$ is not equal to 0 and either $\mathrm{c}(q) = 0$ or $\mathrm{c}(q)$ is not equal to 0. Then Product holds for A, G, p and q.

Step_byP. Let N be a natural number such that Direct_Rec holds for A, G and N. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = N + 1$ and either $\mathrm{c}(p) = 0$ or $\mathrm{c}(p)$ is not equal to 0. Then Product holds for A, G, p and q.

Step. Let N be a natural number such that Direct_Rec holds for A, G and N. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = N + 1$. Then Product holds for A, G, p and q.

At_zero. Let p and q be elements of $\mathcal{P}$ such that $|p| + |q| = 0$. Then Product holds for A, G, p and q.

Direct_induction. Let N be a natural number. Then Direct_Rec holds for A, G and N.

Product. Let p and q be elements of $\mathcal{P}$. Then Product holds for A, G, p and q.

Multiply_common_shift_common. Let c and d be elements of R. Let p and q be elements of $\mathcal{P}$ such that cd is a common divisor of the coefficients of pq. Then cd is a common divisor of the coefficients of $0 + X(pq)$.

Multiply_common. Let c and d be elements of R. Let p and q be elements of $\mathcal{P}$ such that c is a common divisor of the coefficients of p and d is a common

divisor of the coefficients of q. Then cd is a common divisor of the coefficients of pq.

Content_mul_left. Let p and q be elements of $\mathcal{P}$. Then c((pq)) divides c(p)c(q).

Content_mul_right. Let p and q be elements of $\mathcal{P}$. Then c(p)c(q) divides c((pq)).

Direct_gauss. Let p and q be elements of $\mathcal{P}$ such that the function that maps k to p_k is primitive. Let d be an element of R such that d is a common divisor of the coefficients of pq. Let i be a natural number. Then d divides q_i.

15.20 Gauss.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Gauss_list. Let p and q be elements of $\mathcal{P}$ such that the function that maps k to p_k is primitive. Let d be an element of R such that for all natural numbers k, d divides $(pq)_k$. Let j be a natural number. Then d divides q_j.

15.21 Gauss.Over.GaussAt

In this section: Let d be an element of R such that d is not equal to 0.

Definition. Let y be an element of $A[X]$. Then Concl holds for y, if for all natural numbers j, d divides y_j.

Concl_prop. Let y be an element of $A[X]$. Then Concl applied to A, G, d, hd0 and y is a proposition.

Definition. Let x and y be elements of $A[X]$. Then Stmt holds for x and y, if if PolyBase_Over_Prim holds for A, G and x, then if for all natural numbers k, d divides $(xy)_k$, then Concl holds for A, G, d, hd0 and y.

Stmt_prop. Let x and y be elements of $A[X]$. Then Stmt applied to A, G, d, hd0, x and y is a proposition.

On_lists. Let p and q be elements of $\mathcal{P}$. Then Stmt holds for A, G, d, hd0, [p] and [q].

GaussAt_gauss. Let x and y be elements of $A[X]$. Then Stmt holds for A, G, d, hd0, x and y.

15.22 Gauss.Over.DivPrim

In this section: Let E, f and q be elements of $A[X]$ such that PolyBase_Over_Prim holds for A, G and E. Let lam be an element of R such that lam is not equal to 0 and $\mathrm{sc}(A, G, \mathrm{lam}, f) = Eq$.

Dq. Divides holds for A, G, lam, hl and q.

Definition. q' is an element of $A[X]$.

Same. $\mathrm{sc}(A, G, \mathrm{lam}, f) = \mathrm{sc}(A, G, \mathrm{lam}, E\mathrm{q}'(A, G, E, f, q, \mathrm{pE}, \mathrm{lam}, \mathrm{hl}, e))$.

DivPrim_result. E divides f.

15.23 Elimination.Ident

In this section: Let S be a commutative ring.

Ident. Let L, Bt, T, f and g be elements of R. Then $(L\mathrm{Bt})f = (L((\mathrm{Bt}f) + -(Tg))) + (T(Lg))$.

Ident2. Let L, Bt and g be elements of R. Then $(L\mathrm{Bt})g = \mathrm{Bt}(Lg)$.

15.24 Elimination.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let n be a natural number. Let x be an element of $A[X]$. Then x is bounded by n, if Coefficients_Over_Bounded holds for A, GcdDomain_set applied to A and G, n and x.

Elim. Let f, g and gd be elements of $A[X]$. Let lam be an element of R such that lam is not equal to 0, $\mathrm{gd} \mid \mathrm{sc}(A, G, \mathrm{lam}, f)$ and $\mathrm{gd} \mid \mathrm{sc}(A, G, \mathrm{lam}, g)$. Assume that for all elements h of $A[X]$, if h divides f and h divides g, then h divides gd. Then Structure.

Definition. Let f, g and gd be elements of $A[X]$. Let lam be an element of R such that lam is not equal to 0, $\mathrm{gd} \mid \mathrm{sc}(A, G, \mathrm{lam}, f)$ and $\mathrm{gd} \mid \mathrm{sc}(A, G, \mathrm{lam}, g)$. Assume that for all elements h of $A[X]$, if h divides f and h divides g, then h divides gd. Then mkElim applied to f, g, gd, lam, $\mathrm{lam} - \mathrm{nz}$, $\mathrm{div} - \mathrm{f}$, $\mathrm{div} - \mathrm{g}$ and common is an element of Elim applied to A, G, f and g.

Definition. Let f and g be elements of $A[X]$. Let e be an element of Elim applied to A, G, f and g. Then Over_swap applied to f, g and e is an element of Elim applied to A, G, g and f.

Sc_one. Let x be an element of $A[X]$. Then $\mathrm{sc}(A, G, 1, x) = x$.

Definition. Let f and g be elements of $A[X]$ such that f is bounded by 0. Then zero_left applied to f, g and hf is an element of Elim applied to A, G, f and g.

Plus_assoc. Let a, b and c be natural numbers. Then $a + b + c = a + (b + c)$.

Definition. Let x be a natural number. Let x_1 be an element of $A[X]$. Then xp applied to x and x_1 is an element of $A[X]$.

Coef_xp. Let k and i be natural numbers. Let x be an element of $A[X]$. Then $\mathrm{xp}(A, G, k, x)_{(k+i)} = x_i$.

Xp_mul. Let k be a natural number. Let x be an element of $A[X]$. Then $\mathrm{xp}(A, G, k, x) = \mathrm{xp}(A, G, k, 1)x$.

Bound_down. Let x be an element of $A[X]$. Let n be a natural number such that x is bounded by $n + 1$ and $x_n = 0$. Then x is bounded by n.

Definition. Let N be a natural number. Then an element of Over_Rec applied to N is an element of the function that maps N to PiType applied to $A[X]$ and the function that maps f to PiType applied to $A[X]$ and the function that maps g to PiType applied to $\mathbb{N}$ and the function that maps nf to PiType applied to $\mathbb{N}$ and the function that maps ng to PiType applied to $(\mathrm{nf} + \mathrm{ng}) = N$ and the function that maps x to PiType applied to the proposition that f is bounded by nf and the function that maps x_1 to PiType applied to the proposition that g is bounded by ng and the function that maps x_2 to Elim applied to A, G, f and g.

15.25 Elimination.Over.Reduce

In this section: Let f and g be elements of $A[X]$. Let k, nf', ng' and N be natural numbers such that nf' is equal to $k + \mathrm{ng}'$, f is bounded by $\mathrm{nf}' + 1$ and g is bounded by $\mathrm{ng}' + 1$. Let r be an element of Over_Rec applied to A, G and N. Assume that $\mathrm{nf}' + \mathrm{ng}' + 1 = N$. Assume that $g_{\mathrm{ng}'}$ is not equal to 0.

Definition. Reduce_alpha is an element of R.

Definition. Reduce_beta is an element of R.

Definition. Reduce_T is an element of $A[X]$.

Definition. f' is an element of $A[X]$.

Coef_T. Let j be a natural number. Then $(\mathrm{Reduce_T}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})g)_j = \mathrm{Reduce_alpha}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})(\mathrm{xp}(A, G, k, g)_j)$.

Coef_f'. Let j be a natural number. Then $\mathrm{f}'(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})_j =$
$(\mathrm{Reduce_beta}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})(f_j)) + -(\mathrm{Reduce_alpha}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}$

Idx. Let i be a natural number. Then $\mathrm{xp}(A, G, k, g)_{(\mathrm{nf}'+i)} = g_{(\mathrm{ng}'+i)}$.

Top. $\mathrm{f}'(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})_{(\mathrm{nf}'+0)} = 0$.

Above. Let j be a natural number. Then $\mathrm{f}'(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})_{(\mathrm{nf}'+j+1)} = 0$.

Bounded. f′ applied to A, G, f, g, k, nf′, ng′, e, hf, hg, N, r, eN and bn is bounded by nf′.

Common'. Let h be an element of $A[X]$ such that h divides f and h divides g. Then $h \mid \mathrm{f}'(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})$.

Definition. rec is an element of Elim applied to A, G, f′ applied to A, G, f, g, k, nf′, ng′, e, hf, hg, N, r, eN and bn and g.

Definition. Reduce_lam is an element of R.

Sc_f. $\mathrm{sc}(A, G, \mathrm{Reduce_lam}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn}), f) = \mathrm{sc}(A, G, \mathrm{Elim_lam}(A, G, \mathrm{f}'(A, G,$ (Reduce_T$(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn})\mathrm{sc}(A, G, \mathrm{Elim_lam}(A, G, \mathrm{f}'(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{h}$

Sc_g. $\mathrm{sc}(A, G, \mathrm{Reduce_lam}(A, G, f, g, k, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}, N, r, \mathrm{eN}, \mathrm{bn}), g) = \mathrm{constant}(A, \mathrm{GcdDomain_set}(A,$

Definition. Reduce_result is an element of Elim applied to A, G, f and g.

15.26 Elimination.Over.Tops

In this section: Let N, nf′ and ng′ be natural numbers. Let r be an element of Over_Rec applied to A, G and N. Let f and g be elements of $A[X]$. Assume that $\mathrm{nf}' + 1 + \mathrm{ng}' + 1 = N + 1$. Assume that f is bounded by $\mathrm{nf}' + 1$. Assume that g is bounded by $\mathrm{ng}' + 1$.

Definition. Tops_alpha is an element of R.

Definition. Tops_beta is an element of R.

EF. $\mathrm{nf}' + \mathrm{ng}' + 1 = N$.

EG. $\mathrm{nf}' + 1 + \mathrm{ng}' = N$.

ESym. $\mathrm{ng}' + \mathrm{nf}' + 1 = N$.

Definition. Assume that it is not the case that $\mathrm{Tops_alpha}(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$. Assume that it is not the case that $\mathrm{Tops_beta}(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$. Assume that there exists a natural number d such that $\mathrm{nf}' = \mathrm{ng}' + d$ or there exists a natural number d such that $\mathrm{ng}' = \mathrm{nf}' + d$. Then byCmp applied to an, bn and x is an element of Elim applied to A, G, f and g.

Definition. Assume that it is not the case that $\mathrm{Tops_alpha}(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) =$

0. Assume that Tops_beta$(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$ or it is not the case that Tops_beta$(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$. Then Tops_byG applied to an and x is an element of Elim applied to A, G, f and g.

Definition. Assume that Tops_alpha$(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$ or it is not the case that Tops_alpha$(A, G, N, r, f, g, \mathrm{nf}', \mathrm{ng}', e, \mathrm{hf}, \mathrm{hg}) = 0$. Then Tops_byF applied to x is an element of Elim applied to A, G, f and g.

Definition. Tops_result is an element of Elim applied to A, G, f and g.

15.27 Elimination.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let N, nf$'$ and ng be natural numbers. Let r be an element of Over_Rec applied to A, G and N. Let f and g be elements of $A[X]$ such that f is bounded by $\mathrm{nf}' + 1$. Assume that $\mathrm{nf}' + 1 + \mathrm{ng} = N + 1$. Assume that g is bounded by ng. Then onG applied to N, r, f, g, nf$'$, hf, ng, x and x_1 is an element of Elim applied to A, G, f and g.

Definition. Let N, ng and nf be natural numbers. Let r be an element of Over_Rec applied to A, G and N. Let f and g be elements of $A[X]$ such that g is bounded by ng. Assume that $\mathrm{nf} + \mathrm{ng} = N + 1$. Assume that f is bounded by nf. Then onF applied to N, r, f, g, ng, hg, nf, x and x_1 is an element of Elim applied to A, G, f and g.

Definition. Let f and g be elements of $A[X]$. Let ng and nf be natural numbers such that g is bounded by ng, $\mathrm{nf} + \mathrm{ng} = 0$ and f is bounded by nf. Then onZero applied to f, g, ng, hg, nf, x and x_1 is an element of Elim applied to A, G, f and g.

Definition. Let N be a natural number. Then elim applied to N is an element of Over_Rec applied to A, G and N.

Definition. Let f and g be elements of $A[X]$. Let nf and ng be natural numbers such that f is bounded by nf and g is bounded by ng. Then elimination applied to f, g, nf, ng, hf and hg is an element of Elim applied to A, G, f and g.

15.28 Degree.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let x be a natural number. Let x_1 be a function from natural numbers x_1 to elements of R. Then the elements of TopF applied to x, x_1 and x_1 are a basic type.

Definition. Let e be a function from natural numbers x to elements of R. Then an element of DegF applied to x and e is an element of the function that maps e and e to SigmaType applied to $\mathbb{N}$, the function that maps n to SigmaType applied to Bnd applied to A, G, n and x and the function that maps x_1 to TopF applied to A, G, n and x and e.

Top_eq. Let u and v be elements of Ring_Top. Then $u = v$.

Top_prop. Let n be a natural number. Let e be a function from natural numbers x to elements of R. Then TopF applied to A, G, n and e is a proposition.

Bnd_prop. Let n be a natural number. Let e be a function from natural numbers x to elements of R. Then Bnd applied to A, G, n and e is a proposition.

Top_vs_bound. Let m, d and n be natural numbers. Let e be a function from natural numbers x to elements of R such that Bnd holds for A, G, m and e. Let x be an element of TopF applied to A, G, n and e. Assume that $n = m + d + 1$. Then we have a contradiction.

Unique_step. Let e be a function from natural numbers x to elements of R. Let n, m and d be natural numbers. Let tn be an element of TopF applied to A, G, n and e. Assume that Bnd holds for A, G, m and e. Assume that $n = m + d$. Then $n = m$.

Unique_cmp. Let e be a function from natural numbers x to elements of R. Let n and m be natural numbers such that Bnd holds for A, G, n and e. Let tn be an element of TopF applied to A, G, n and e. Assume that Bnd holds for A, G, m and e. Let tm be an element of TopF applied to A, G, m and e. Assume that there exists a natural number d such that $n = m + d$ or there exists a natural number d such that $m = n + d$. Then $n = m$.

Deg_prop. Let e be a function from natural numbers x to elements of R. Then DegF applied to A, G and e is a proposition.

Zero_cons. Let c be an element of R. Let p be an element of $\mathcal{P}$ such that Bnd holds for A, G, 0 and the function that maps k to p_k. Assume that $c = 0$. Then Bnd holds for A, G, 0 and the function that maps k to $(c + Xp)_k$.

Definition. Let c be an element of R. Let p be an element of $\mathcal{P}$ such that Bnd holds for A, G, 0 and the function that maps k to p_k. Assume that $c = 0$ or c is not equal to 0. Then byC applied to c, p, b and x is an element of DegF applied to A, G and the function that maps k to $(c + Xp)_k$.

Definition. Let c be an element of R. Let p be an element of $\mathcal{P}$. Let n be a natural number such that Bnd holds for A, G, n and the function that maps k to p_k. Let x_1 be an element of TopF applied to A, G, n and the function that maps k to p_k. Then cons_deg applied to c, p, n, x and x_1 is an element of DegF applied to A, G and the function that maps k to $(c + Xp)_k$.

Definition. Let p be an element of $\mathcal{P}$. Then deg_list applied to p is an element of DegF applied to A, G and the function that maps k to p_k.

Definition. Let x be an element of $A[X]$. Then deg applied to x is an element of DegF applied to A, G and the function that maps k to x_k.

Definition. Let x be an element of $A[X]$. Then Over_dg applied to x is a natural number.

Dg_bnd. Let x be an element of $A[X]$. Then Bnd holds for A, G, Over_dg applied to A, G and x and the function that maps k to x_k.

Zero_of_coefs. Let x be an element of $A[X]$ such that for all natural numbers i, $x_i = 0$. Then $x = 0$.

15.29 PolyGcd.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Definition. Let A be a commutative ring. Let G be a gcd domain on A. Let x be an element of $A[X]$. Then $\mathrm{c}(x)$ is an element of R.

Zero_content. Let x be an element of $A[X]$ such that $\mathrm{c}(x) = 0$. Then $x = 0$.

Content_zero. Let x be an element of $A[X]$ such that for all natural numbers i, $x_i = 0$. Then $\mathrm{c}(x) = 0$.

Zero_coef. Let x be an element of $A[X]$ such that $x = 0$. Let i be a natural number. Then $x_i = 0$.

Zero_mul_coef. Let h and r be elements of $A[X]$ such that $h = 0$. Let i be a natural number. Then $(hr)_i = 0$.

Nz_div. Let h and f be elements of $A[X]$ such that h divides f and $\mathrm{c}(f)$ is not equal to 0. Then $\mathrm{c}(h)$ is not equal to 0.

Shuffle. Let c be an element of R. Let E and q be elements of $A[X]$. Then $\mathrm{sc}(A, G, c, E)q = E\mathrm{sc}(A, G, c, q)$.

15.30 PolyGcd.Over.PP

In this section: Let x be an element of $A[X]$ such that $\mathrm{c}(x)$ is not equal to 0.

Definition. PP_c is an element of R.

Definition. xs is an element of $A[X]$.

PP_eq. $x = \mathrm{sc}(A, G, \mathrm{PP_c}(A, G, x, \mathrm{hc}), \mathrm{xs}(A, G, x, \mathrm{hc}))$.

PP_prim. PolyBase_Over_Prim holds for A, G and xs applied to A, G, x and hc.

15.31 PolyGcd.Over.Main

In this section: Let f and g be elements of $A[X]$ such that $\mathrm{c}(f)$ is not equal to 0 and $\mathrm{c}(g)$ is not equal to 0.

Definition. el is an element of Elim applied to A, G, f and g.

Definition. Main_lam is an element of R.

ND. $\mathrm{c}(\mathrm{gd}(A, G, f, g, \mathrm{el}(A, G, f, g, \mathrm{nf}, \mathrm{ng})))$ is not equal to 0.

Definition. E is an element of $A[X]$.

Definition. cD is an element of R.

Main_ef. $\mathrm{sc}(A, G, \mathrm{Main_lam}(A, G, f, g, \mathrm{nf}, \mathrm{ng}), f) = E(A, G, f, g, \mathrm{nf}, \mathrm{ng})\mathrm{sc}(A, G, \mathrm{cD}(A, G, f, g, \mathrm{nf}, \mathrm{ng}), \pi_1(\mathrm{div}$

Eg. $\mathrm{sc}(A, G, \mathrm{Main_lam}(A, G, f, g, \mathrm{nf}, \mathrm{ng}), g) = E(A, G, f, g, \mathrm{nf}, \mathrm{ng})\mathrm{sc}(A, G, \mathrm{cD}(A, G, f, g, \mathrm{nf}, \mathrm{ng}), \pi_1(\mathrm{div_g}(A,$

Ef. $E(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid f$.

Eg. $E(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid g$.

Definition. gam is an element of R.

Ngam. It is not the case that $\mathrm{gam}(A, G, f, g, \mathrm{nf}, \mathrm{ng}) = 0$.

Definition. Gp is an element of $A[X]$.

15.32 PolyGcd.Over.Main.Part

In this section: Let x be an element of $A[X]$ such that $E(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid x$ and $\mathrm{gam}(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid \mathrm{c}(x)$.

Dgi. Let i be a natural number. Then $\mathrm{gam}(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid (E(A, G, f, g, \mathrm{nf}, \mathrm{ng})\pi_1(\mathrm{dE}))_i$.

S. Quot holds for A, G, gam applied to A, G, f, g, nf and ng, ngam applied to A, G, f, g, nf and ng and $\pi_1(\mathrm{dE})$.

Part_result. $\mathrm{Gp}(A, G, f, g, \mathrm{nf}, \mathrm{ng}) \mid x$.

15.33 PolyGcd.Over.Main.Common

In this section: Let h be an element of $A[X]$ such that h divides f and h divides g.

HD. $h \mid \mathrm{gd}(A, G, f, g, \mathrm{el}(A, G, f, g, \mathrm{nf}, \mathrm{ng}))$.

Nh. $\mathrm{c}(h)$ is not equal to 0.

Definition. ch is an element of R.

Ch_div. Let x be an element of $A[X]$ such that h divides x. Let i be a natural number. Then $\mathrm{ch}(A, G, f, g, \mathrm{nf}, \mathrm{ng}, h, \mathrm{df}, \mathrm{dg}) \mid x_i$.

Ch_gam. $\mathrm{ch}(A, G, f, g, \mathrm{nf}, \mathrm{ng}, h, \mathrm{df}, \mathrm{dg}) \mid \mathrm{gam}(A, G, f, g, \mathrm{nf}, \mathrm{ng})$.

ED. $\mathrm{sc}(A, G, \mathrm{cD}(A, G, f, g, \mathrm{nf}, \mathrm{ng}), E(A, G, f, g, \mathrm{nf}, \mathrm{ng})) = \mathrm{xs}(A, G, h, \mathrm{nh}(A, G, f, g, \mathrm{nf}, \mathrm{ng}, h, \mathrm{df}, \mathrm{dg}))\mathrm{sc}(A, G,$

HsE. $\mathrm{xs}(A, G, h, \mathrm{nh}(A, G, f, g, \mathrm{nf}, \mathrm{ng}, h, \mathrm{df}, \mathrm{dg})) \mid E(A, G, f, g, \mathrm{nf}, \mathrm{ng})$.

CC. $\mathrm{constant}(A, \mathrm{GcdDomain_set}(A, G), \mathrm{ch}(A, G, f, g, \mathrm{nf}, \mathrm{ng}, h, \mathrm{df}, \mathrm{dg})) \mid \mathrm{constant}(A, \mathrm{GcdDomain_set}(A, G),$

Common_result. $h \mid \mathrm{Gp}(A, G, f, g, \mathrm{nf}, \mathrm{ng})$.

15.34 PolyGcd.Over.Main

In this section: Let f and g be elements of $A[X]$ such that $\mathrm{c}(f)$ is not equal to 0 and $\mathrm{c}(g)$ is not equal to 0.

GcdOf. f and g have a greatest common divisor.

15.35 PolyGcd.Over

In this section: Let A be a commutative ring. Let G be a gcd domain on A.

Over_byG. Let f and g be elements of $A[X]$ such that $\mathrm{c}(f)$ is not equal to 0 and either $\mathrm{c}(g) = 0$ or $\mathrm{c}(g)$ is not equal to 0. Then f and g have a greatest common divisor.

Over_byF. Let f and g be elements of $A[X]$ such that $\mathrm{c}(f) = 0$ or $\mathrm{c}(f)$ is not equal to 0. Then f and g have a greatest common divisor.

GcdP. Let f and g be elements of $A[X]$. Then f and g have a greatest common divisor.

ByDec. Let x be an element of $A[X]$ such that $\mathrm{c}(x) = 0$ or $\mathrm{c}(x)$ is not equal to 0. Then $x = 0$ or x is not equal to 0.

DecP. Let x be an element of $A[X]$. Then $x = 0$ or x is not equal to 0.

NontrivialP. 1 is not equal to 0.

List_content_nz. Let p be an element of $\mathcal{P}$ such that $[p]$ is not equal to 0. Then $\mathrm{c}(p)$ is not equal to 0.

Integral_lists. Let p and q be elements of $\mathcal{P}$ such that $[p]$ is not equal to 0, $[q]$ is not equal to 0 and $[p][q] = 0$. Then we have a contradiction.

Definition. Let x and y be elements of $A[X]$. Then Integral holds for x and y, if if x is not equal to 0, then if y is not equal to 0, then xy is not equal to 0.

IntegralP. Let x and y be elements of $A[X]$. Then Integral holds for A, G, x and y.

Definition. Over_domain is a gcd domain on $A[X]$.